Industrial Policies Cascade in Production Networks

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This version: June, 2026

Abstract

This paper studies industrial policy in production networks with imperfect competition and firm entry and exit. The first-order output response to a sectoral subsidy decomposes into two sufficient statistics: *distortion centrality*, the intensive-margin gain from improving allocative efficiency; and *entry centrality*, the extensive-margin value of the cascading firm entry a subsidy triggers across the network. Driven by returns to variety and pro-competitive markup responses, entry centrality re-ranks sectors toward downstream industries whose expansion supports more entry along their supply chains. Our quantitative analysis shows that entry centrality is negatively correlated with distortion centrality, and reduced-form evidence from China links interventions to both statistics.

Keywords: production networks, industrial policy, firm entry, returns to variety, variable markups, sufficient statistics

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1 Introduction

Industrial policy has returned to the center of economic policymaking, from the U.S. CHIPS Act and Inflation Reduction Act to the European Union's green-industrial agenda and China's long-running programs of sector-specific support. This revival sharpens an old question: which sectors generate the largest aggregate return to targeted intervention? Subsidies may reduce wedges; but when markets are imperfectly competitive and firms enter, they also change market size, the number of firms and varieties, and, under variable markups, the competitive pressure those firms face. The classic "linkages" tradition emphasizes sectors with dense input–output connections (Hirschman, 1958). Liu (2019) formalizes this logic in a distorted input–output economy, showing that *distortion centrality*—a statistic that compounds wedges along the supply chain—ranks sectors by the first-order output gain from subsidizing them. Yet much observed policy targets large downstream markets rather than upstream inputs. Automobiles are a textbook object of industrial policy, but they are not an upstream input sector—and the same is true of transportation equipment, housing, and construction, which are frequent targets of industrial policy or other large-scale sectoral intervention despite sitting well below the top of the network's upstream ranking. Because distortion centrality is largest upstream, where wedges compound as they pass through to downstream users, a criterion built on this intensive margin does not naturally explain why such downstream sectors are so often targeted. Their relevance becomes intelligible once a subsidy can expand a sector's output and the demand it places on its suppliers, setting off firm entry that cascades backward through the production network.

What this criterion leaves out is the extensive margin. Distortion centrality holds the mass of entrants fixed: as in much of the production-network literature (Baqaee and Farhi, 2019, 2020b), subsidies operate through an intensive margin, reducing wedges and reallocating resources across a given production network. But policy also moves the number of firms. When firms face fixed operating costs and sectoral output aggregates varieties with increasing returns to variety, each sector's mass of firms is pinned by its sales. A subsidy lowers the targeted sector's wedge and raises its demand for inputs, lifting its suppliers' sales and the number of firms they can support; this entry then propagates further upstream. A sectoral subsidy can therefore set off a cascade of firm entry across the production network (Baqaee, 2018; Taschereau-Dumouchel, 2026). A sector may have modest distortion centrality yet generate a large output response because subsidizing it induces entry throughout its supply chain. A sufficient statistic for the output effect must capture not only a policy's direct effect on distortions but also the entry cascade it sets off across sectors.

This paper develops a sufficient-statistic framework for the output effects of sectoral subsidies in such an environment. We embed endogenous firm entry and increasing returns to variety, in the spirit of Dixit and Stiglitz (1977) and Baqaee (2018), into a distorted input–output economy à la Baqaee and Farhi (2020b). Sectoral subsidies lower wedges and change sectoral sales. Through the free-entry condition, these sales changes induce firm entry and exit, which in turn affect sectoral prices and reallocate expenditure across the network. The network is therefore not only a propagation device for productivity or wedge shocks; it is also the channel through which sectoral policy reshapes market size, entry incentives, and competitive pressure across industries.

Our central result decomposes the first-order elasticity of aggregate output with respect to a sectoral subsidy into two network statistics. The first is *distortion centrality*, the intensive-margin gain from reducing wedges while holding the mass of firms fixed, as in Liu (2019). The second, which is new, is *entry centrality*: the extensive-margin output value of the entry cascade the subsidy sets off across the network. Together they form a sufficient statistic for the first-order output response to sectoral policy, and both are computed from observable input–output shares, Domar weights, substitution elasticities, wedges, the strength of returns to variety, and the pro-competitive response of markups to entry. The policy ranking is then no longer governed by wedges alone: a sector is a valuable target when subsidizing it sets off high-output-value entry elsewhere in the network.

The contribution is threefold. First, we turn endogenous entry into a sufficient statistic for sectoral policy, so the extensive margin can be ranked with the same discipline as distortion centrality. Second, we characterize the equilibrium entry response in closed form as a network multiplier, driven by returns to variety and—under variable markups—a pro-competitive response governed by the curvature of demand. Third, we show quantitatively that the extensive margin changes which sectors look attractive for policy, shifting attention from upstream distorted inputs toward sectors whose subsidy sets off a large entry cascade along their supply chains. Because entry centrality is measured per unit of sales, the criterion is the entry a subsidy mobilizes upstream, not a market's size; in our U.S. calibration these sectors are typically downstream and consumer-facing—markets that observed policy often targets but that an intensive-margin criterion overlooks. The synthesis is itself the contribution: neither an intensive-margin statistic (Liu, 2019) nor a network entry cascade (Baqaee, 2018) alone ranks sectors by the output value of subsidy-induced entry, and bringing them together *reorders* the policy ranking rather than refining it—entry centrality is negatively correlated with distortion centrality in U.S. data and concentrated downstream rather than upstream.

The entry cascade has a simple logic. A subsidy lowers a sector's price and raises the derived demand for its suppliers, stimulating entry backward along the supply chain; the induced relative-price changes simultaneously reallocate expenditure and shift entry incentives across sectors. Because each sector's mass of firms is pinned by the sales it can sustain, the economy-wide entry response solves a network fixed point that we characterize in closed form. With Kimball demand, entry also has a pro-competitive effect: by reducing each producer's relative scale, it makes residual demand more elastic and compresses desired markups.

The framework clarifies why entry centrality is distinct from distortion centrality. In the Cobb–Douglas, Dixit–Stiglitz benchmark, the entry component is governed by a backward supply-chain multiplier. This multiplier measures the total direct and indirect upstream production activated by expanding a sector. Entry centrality is therefore high for sectors whose expansion pulls strongly on their suppliers, not necessarily for sectors that are upstream or highly distorted themselves. The channel is nonetheless a market-power phenomenon: the operating margin and the returns to variety finance the entry response, and as firms approach perfect competition, both vanish and entry ceases to move output.

We illustrate the decomposition in three simple economies. In a roundabout economy, both distortion centrality and entry centrality are positive when firms price above marginal cost and use intermediate inputs, but the entry component becomes more important as production becomes more responsive to relative prices. In a horizontal economy with homogeneous wedges, distortion centrality is zero, while entry centrality depends on whether the subsidy reallocates expenditure toward sectors with stronger or weaker returns-to-variety effects. In a vertical supply chain, resource allocation is pinned down by technology, so distortion centrality is again zero; nevertheless, a subsidy to a downstream sector stimulates entry in all upstream sectors. These examples show that entry centrality is not a relabeling of upstreamness or distortion centrality. It captures a separate extensive-margin propagation mechanism.

We then quantify the two statistics using U.S. input–output data. The two measures are far from collinear: they are in fact negatively correlated across sectors, and a regression of total centrality on distortion centrality alone leaves more than a third of the variation unexplained. They also occupy opposite ends of the supply chain. Distortion centrality is strongly and positively correlated with sectoral upstreamness, reproducing the forward-propagation logic of Liu (2019), whereas entry centrality is strongly negatively correlated with it: the extensive-margin return to a subsidy is largest in downstream, consumer-facing sectors such as housing, food and beverage, and motor vehicles, and smallest in upstream raw-materials and business-service sectors. A ranking based on distortion centrality alone therefore understates the output return to subsidies that mobilize entry along a sector's supply chain—in our calibration, these are downstream markets, but they are selected by the entry a subsidy sets off rather than by their size.

We also provide reduced-form evidence from China. Using China's 2007 input–output table and sectoral intervention measures from Liu (2019), we ask whether observed sectoral policies are correlated with the two centrality measures. The results are descriptive rather than causal. They show that private firms in sectors with higher distortion centrality and higher entry centrality tend to face more favorable financial conditions, including lower effective interest rates and higher debt ratios. Adding entry centrality substantially increases explanatory power, especially for effective interest rates, and the relationship is robust to controlling for capital intensity, Lerner indices, entry-scale proxies, and export intensity. The evidence is therefore consistent with the view that observed interventions load not only on pre-existing distortions, but also on sectors whose subsidies are associated with stronger entry spillovers through the production network.

Throughout the paper, our object is the aggregate output or productivity response to sectoral policy, not a complete welfare criterion. This distinction matters because free entry in monopolistic-competition models can be socially excessive or insufficient depending on the structure of preferences, fixed costs, and business-stealing effects (Mankiw and Whinston, 1986; Dhingra and Morrow, 2019); Baqaee and Farhi (2020a) characterize this welfare margin directly through the social cost of distortions, from which we abstract. Our sufficient statistics measure how subsidies affect real output through intensive and extensive margins. Thus the analysis should be read as an output-based industrial-policy criterion, not as a claim that all policy-induced entry is welfare improving. Incorporating the full welfare tradeoff associated with entry is an important extension, but it is not required for the output decomposition developed here.

Related Literature. The closest antecedent is Liu (2019), who gives a formal network rationale for industrial policy in an economy with pre-existing imperfections. In his framework, distortion centrality is a sufficient statistic for the first-order gain from subsidizing a sector, and the logic favors sectors whose wedges propagate through backward demand linkages. We keep this policy question and this sufficient-statistic discipline, but change the margin on which policy operates. In our economy, a sectoral subsidy changes not only wedges and resource allocation holding products fixed, but also the mass of firms and varieties. Distortion centrality is therefore the intensive-margin component of the output response, while entry centrality measures the output value of the induced entry cascade.

The paper also contributes to work on production networks, aggregation, and misallocation. The benchmark network literature studies how microeconomic shocks propagate to aggregates through input–output linkages (Long and Plosser, 1983; Gabaix, 2011; Acemoglu et al., 2012; Carvalho and Tahbaz-Salehi, 2019). With distortions, wedges and reallocation become first-order forces, as in Jones (2011), Bigio and La'o (2020), and the general aggregation framework of Baqaee and Farhi (2020b). This connects our analysis to the broader misallocation literature (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Midrigan and Xu, 2014). Relative to these papers, we add an endogenous extensive margin: sectoral sales determine entry, entry changes prices and wedges, and those changes feed back through the production network.

Our entry mechanism builds most directly on Baqaee (2018), who shows that entry and exit amplify shocks in production networks through cascades of product creation and destruction, and it relates to work on endogenous entry and product variety in macroeconomics (Bilbiie et al., 2012, 2019) and on supplier relationships and endogenous production networks (Acemoglu and Azar, 2020; Carvalho et al., 2021; Elliott et al., 2022; Baqaee et al., 2025; Taschereau-Dumouchel, 2026). The closest theoretical framework is Baqaee and Farhi (2020a), who study aggregation with entry, economies of scale, and distortions and use it to characterize the social cost of distortions and optimal policy. We work in the same general environment—an input–output network with distortions, firm entry, and returns to variety—and broaden it to variable markups through Kimball demand, so that entry also compresses markups. Our question and results differ: we summarize the *positive* first-order output response to a sectoral subsidy in two observable network statistics, distortion and entry centrality, rather than the social cost of distortions or optimal policy, and we characterize the equilibrium entry response in closed form as a network multiplier. The two analyses also point in different directions: their welfare-optimal intervention favors upstream sectors with strong scale economies, whereas entry centrality is largest downstream, because subsidizing a downstream sector mobilizes those upstream scale economies through the backward entry cascade. We therefore read the entry cascade as a policy statistic rather than only a shock-propagation mechanism.

The paper is also related to the renewed literature on industrial policy (Hirschman, 1958; Song et al., 2011; Aghion et al., 2015; Itskhoki and Moll, 2019; Juhász et al., 2024). Bartelme et al. (2025) develop a case for industrial policy based on cross-sector scale economies. Our scale economies instead arise from firm entry and returns to variety, so the relevant scale effect is not only a sector's own scale elasticity but also the network-wide entry response it triggers. The reduced-form evidence from China connects to work studying Chinese industrial transformation, policy, and firm performance, including Brandt et al. (2012), Khandelwal et al. (2013), Liu (2019), and Chen et al. (2026); our empirical exercise is descriptive and asks whether observed interventions load on the two sufficient statistics highlighted by the model.

Finally, the pro-competitive channel links the paper to industrial organization and trade work on markups, competition, and variable demand elasticities. Our Kimball structure builds on Kimball (1995) and the demand systems studied by Zhelobodko et al. (2012), Mrázová and Neary (2017), and Mrázová and Neary (2020). The mechanism is especially close to the IO/trade literature showing that competition and market size shape markups and productivity, including Syverson (2004), Atkeson and Burstein (2008), De Loecker and Warzynski (2012), Arkolakis et al. (2019), and Aghion et al. (2005). We are particularly related to the work of Edmond et al. (2015) and Edmond et al. (2023), who quantify how markup dispersion and pro-competitive forces affect the gains from trade and the aggregate cost of market power. Our pro-competitive elasticity summarizes a related response, but the policy experiment is different: rather than studying trade liberalization or the static cost of markups, we ask how entry induced by a sectoral subsidy compresses wedges and propagates through input–output linkages. Here the network plays the organizing role: sectoral policy changes the competitive environment, and input–output linkages determine where that pro-competitive response is valuable. Methodologically, this also connects to Baqaee et al. (2024), who use nonparametric Kimball demand to summarize how wedges respond to cost and demand shocks; here the same demand-side discipline is redirected to entry and sectoral industrial policy. Because the main text evaluates subsidies by their effect on aggregate output, we abstract from the classic welfare question of whether entry is socially excessive or insufficient (Spence, 1976; Mankiw and Whinston, 1986; Dhingra and Morrow, 2019).

Outline. The rest of the paper is organized as follows. Section 2 introduces the model. Section 3 derives the aggregation and propagation results and illustrates the decomposition through examples. Section 4 quantifies distortion centrality and entry centrality in U.S. data and presents reduced-form evidence from China. Section 5 concludes.

2 Framework

This section lays out the economic environment and the notation used throughout the paper. The economy consists of monopolistically competitive sectors connected through input–output linkages. Firms produce differentiated varieties, enter subject to fixed overhead costs, and face Kimball demand within each sector, so entry affects both sectoral price indexes and desired markups. The key equilibrium object is the free-entry condition, which links the mass of firms in a sector to its market size and operating surplus. We then introduce the input–output objects and sector-level primitives used to characterize how shocks and policies propagate through the economy.

2.1 Model Environment

We study a static production-network economy with N sectors, monopolistically competitive producers, and endogenous firm entry. Firms use labor and intermediate inputs to produce differentiated varieties, which are aggregated into sectoral goods by homothetic Kimball aggregators. Final uses are aggregated into real GDP by a constant-returns aggregator. Sector-specific policy interventions distort producers' marginal costs and are financed by lump-sum taxes on final users. The key departure from the standard Dixit–Stiglitz environment is that Kimball demand makes residual demand elasticities depend on firms' relative scale. As a result, entry affects aggregate output through two channels: a returns-to-variety channel, which lowers sectoral price indexes for given variety prices, and a pro-competitive channel, which compresses desired markups by making residual demand more elastic. The model can therefore be read as an imperfect-competition model of sectoral policy embedded in an input–output network.

Firms. Within each sector k, firm ν uses labor input $l_k(\nu)$ and sectoral products as intermediate inputs $\{x_{kh}(\nu)\}_{h=1}^N$ to produce a differentiated variety $y_k(\nu)$. The production function of firm ν in sector k is

$$y_k(\nu) = z_k\, F\Big(l_k(\nu), \{x_{kh}(\nu)\}_{h=1}^N\Big),$$

where z_k is a Hicks-neutral productivity shifter common to all firms in sector k, and F is a constant-returns technology.

The sectoral output Y_k aggregates its differentiated varieties through a homothetic Kimball aggregator, defined by

$$\int_0^{M_k} \Upsilon_k(q_k(\nu))\, d\nu = 1, \qquad q_k(\nu) \equiv \frac{y_k(\nu)}{Y_k},$$

where M_k is the mass of firms and Υ_k is increasing and strictly concave. The Dixit and Stiglitz (1977) aggregator is the special case in which Υ_k is a power function, $\Upsilon_k(q_k(\nu)) = q_k(\nu)^{(\varepsilon_k-1)/\varepsilon_k}$.

The ideal sectoral price index is given by

$$P_k = \varrho_k \int_0^{M_k} \Upsilon'_k(q_k(\nu))\, q_k(\nu) d\nu,$$

where ϱ_k is the scalar that normalizes firm-level inverse demands: $p_k(\nu)/\varrho_k = \Upsilon'_k(q_k(\nu))$.

Conditional on entry, all firms optimally choose the combination of inputs to minimize the marginal cost,

$$mc_k = \min_{\{x_{kh}(\nu)\}, l_k(\nu)} W l_k(\nu) + \sum_{h=1}^{N} P_h x_{kh}(\nu),$$

subject to a unit production constraint.

Firms incur a fixed overhead cost f_k to operate, which we write as a fixed fraction ϕ_k of nominal output, $f_k = \phi_k \sum_{j=1}^{N} P_j C_j$, with ϕ_k the structural entry barrier. Expressing the overhead relative to market size makes the equilibrium mass of firms a function of sales *shares*, and hence independent of the price numeraire. We hold ϕ_k fixed throughout the main analysis, so that the mass of entrants responds only through sectoral sales shares. The profits are expressed as

$$\pi_k(\nu) = p_k(\nu) y_k(\nu) - \tau_k^{-1} mc_k y_k(\nu) - f_k$$

where τ_k is the effective subsidy rate provided by the government that discounts the marginal cost of production: $\tau_k > 1$ implies a subsidy reducing costs while $\tau_k < 1$ implies a tax increasing costs. The profit-maximizing price can be written as a desired markup times subsidy-adjusted marginal cost, $p_k(\nu) = \frac{\varepsilon_k}{\varepsilon_k - 1} \tau_k^{-1} mc_k = \mu_k \, mc_k$, where ε_k is the perceived price elasticity of demand. Unlike the CES demand system, which imposes that the price elasticity of demand is constant in the cross section of firms, the Kimball aggregator lets the elasticity facing a firm vary with its relative scale $q_k(\nu)$. Evaluated at the symmetric equilibrium, where $q_k(\nu) = q_k$, this elasticity is

$$\varepsilon_k \equiv - \left. \frac{\partial \log q_k(\nu)}{\partial \log p_k(\nu)} \right|_{\varrho_k} = - \frac{\Upsilon'_k(q_k)}{q_k \, \Upsilon''_k(q_k)}. \tag{1}$$

In the symmetric equilibrium all firms in sector k are identical and operate at the common relative scale q_k solving $M_k \Upsilon_k(q_k) = 1$, so q_k falls with the mass of firms M_k.

Entry changes each producer's relative scale q_k, and with it the sectoral price index and the desired markup; two elasticities summarize these responses: the *returns to variety* η_k—which governs how the sectoral price index responds to entry and is shaped by Υ_k—and the *pro-competitive elasticity* $\xi_k \geq 0$, which governs how the desired markup responds to entry and is shaped by the curvature of Υ_k. The returns to variety capture how the sectoral price responds to firm entry, holding variety prices fixed:

$$\eta_k \equiv - \frac{d \log P_k}{d \log M_k} = \frac{\Upsilon_k(q_k)}{q_k \, \Upsilon'_k(q_k)} - 1. \tag{2}$$

Under the Dixit–Stiglitz aggregator, the desired markup is constant and the returns to variety are the only channel through which entry raises output. Under the Kimball aggregator the desired sectoral markup $\frac{\varepsilon_k}{\varepsilon_k-1}$ is no longer constant: a larger mass of firms makes each producer's residual demand more elastic, raising ε_k and compressing the markup. We summarize this through the pro-competitive elasticity, which is a curvature object that sets the *response* of the wedge to entry:

$$\xi_k \equiv -\frac{d\log\mu_k}{d\log M_k} = -\frac{1}{\varepsilon_k-1}\,\frac{\Upsilon_k(q_k)}{q_k\,\Upsilon_k'(q_k)}\left[\frac{q_k\Upsilon_k''(q_k)}{\Upsilon_k'(q_k)} - \frac{q_k\Upsilon_k'''(q_k)}{\Upsilon_k''(q_k)} - 1\right]. \tag{3}$$

The term in square brackets is the *superelasticity* of demand, $\frac{d\log\varepsilon_k}{d\log q_k}$, which records how the demand elasticity itself moves with scale—and under Marshall's Second Law of Demand this term is nonpositive, so $\xi_k \geq 0$.

Final demand. Let $\mathcal{Y}$ be a constant-returns aggregator over final uses. Real output is the maximal aggregate final use,

$$Y = \max_{\{C_k\}_{k=1}^N} \mathcal{Y}\Big(\{C_k\}_{k=1}^N\Big),$$

subject to the budget constraint

$$\sum_{k=1}^N P_k C_k = W\bar{L} + \sum_{k=1}^N \int_0^{M_k} [\pi_k(\nu) + f_k]\, d\nu - T.$$

Here C_k denotes final use of good k, $W\bar{L}$ is income from the inelastically supplied labor endowment $\bar{L} = 1$, and T is a lump-sum tax. Profits $\pi_k(\nu)$ are net of the overhead, while the overhead f_k is rebated lump-sum to final users. Thus the overhead affects firms' entry decisions but does not absorb aggregate resources.[1] With free entry, $\pi_k(\nu) = 0$ in equilibrium.

Government. Throughout the main analysis, the instrument of industrial policy is a set of sector-specific subsidy rates $\{\tau_k\}_{k=1}^N$ that scale producers' marginal costs from mc_k to $\tau_k^{-1} mc_k$, so that $\tau_k > 1$ is a subsidy and $\tau_k < 1$ is a tax. In the Online Appendix C, we study industrial policy that acts directly on the overhead cost f_k.

The government balances its budget using a lump-sum tax T on final consumers,

$$T = \sum_{k=1}^N \left(1 - \tau_k^{-1}\right) mc_k \int_0^{M_k} y_k(\nu) d\nu.$$

[1] In the Online Appendix D we instead let the overhead be paid in labor, so that it enters the resource constraint, following Dixit and Stiglitz (1977), Melitz (2003), and Baqaee (2018); entry then carries a resource cost and the social optimum becomes interior.

Market clearing. Goods and labor markets clear. Each sector's output is absorbed by final use and by intermediate demand from the sectors that use it as an input, and aggregate labor demand exhausts the endowment:

$$Y_k = C_k + \sum_{h=1}^{N} X_{hk} \quad \forall k, \qquad \sum_{k=1}^{N} L_k = \bar{L},$$

where $X_{hk} = \int_0^{M_h} x_{hk}(\nu)d\nu$ is sector h's total demand for good k and $L_k = \int_0^{M_k} l_k(\nu)d\nu$ is total labor employed in sector k.

Equilibrium. Given productivities z_k, aggregators Υ_k and $\mathcal{Y}$, overhead shares ϕ_k, labor endowment $\bar{L}$, and subsidies τ_k, a general equilibrium is a collection of variety prices $p_k(\nu)$, wage W, intermediate inputs $x_{kh}(\nu)$, labor inputs $l_k(\nu)$, outputs $y_k(\nu)$, final uses C_k, and firm masses M_k such that: (i) firms choose inputs and set prices optimally; (ii) final uses maximize real output subject to the final-use expenditure constraint; (iii) the government budget constraint holds; (iv) goods and labor markets clear; and (v) free entry implies zero profits.

2.2 Definitions and Notation

We now introduce several pieces of notation used throughout the analysis. To write the network objects compactly, we treat labor as an additional industry, indexed by $L = N+1$, interpreted as an endowment producer. Throughout, indexed variables such as μ_k, λ_k, and M_k denote sector-level scalars; bold lowercase symbols such as $\boldsymbol{\mu}$, $\boldsymbol{\lambda}$, and $\boldsymbol{M}$ denote the corresponding vectors; and $\mathrm{diag}(\boldsymbol{\mu})$ denotes the diagonal matrix with vector $\boldsymbol{\mu}$ on the diagonal.

Input–output matrices. Following Liu (2019) and Baqaee and Farhi (2020b), we distinguish between two input–output matrices: a cost-based, or forward, matrix and a revenue-based, or backward, matrix. The cost-based matrix is an $(N+1) \times (N+1)$ matrix whose kh-th element measures the direct exposure of sector k's marginal cost to the price of input h:

$$\tilde{\Omega}_{kh} \equiv \frac{\partial \log mc_k}{\partial \log P_h} = \frac{P_h x_{kh}(\nu)}{mc_k y_k(\nu)}.$$

Here, the first N rows and columns correspond to sectoral output, while the last row and column correspond to labor. Because the production technology is constant returns to scale, input use is proportional to output and marginal cost is common across varieties, so this per-variety cost share equals the ratio of sector k's total expenditure on input h to its total cost:

$$\tilde{\Omega}_{kh} = \frac{\int_0^{M_k} P_h x_{kh}(\nu)d\nu}{\int_0^{M_k} mc_k y_k(\nu)d\nu} = \frac{P_h X_{kh}}{\mu_k^{-1} P_k Y_k}.$$

The second equality uses that each firm prices at the wedge μ_k over marginal cost, $p_k(\nu) = \mu_k mc_k$, so that sectoral sales equal the wedge times sectoral cost, $P_k Y_k = \mu_k mc_k \int_0^{M_k} y_k(\nu) d\nu$.

The revenue-based matrix instead records input expenditures relative to sectoral sales:

$$\Omega_{kh} \equiv \frac{P_h X_{kh}}{P_k Y_k} = \mu_k^{-1} \tilde{\Omega}_{kh},$$

or, in matrix notation, $\Omega \equiv \operatorname{diag}(\boldsymbol{\mu})^{-1}\tilde{\Omega}$, where $\boldsymbol{\mu}$ is the vector of sectoral wedges. The distinction between $\tilde{\Omega}$ and Ω is important. The former describes how input prices propagate forward into marginal costs; the latter describes how sectoral sales are allocated backward across intermediate-input suppliers and primary factors. The two matrices coincide under marginal-cost pricing, but differ whenever sectoral wedges create a gap between revenues and costs.

The corresponding cost-based and revenue-based Leontief inverses are

$$\tilde{\Psi} \equiv (I - \tilde{\Omega})^{-1}, \quad \text{and} \quad \Psi \equiv (I - \Omega)^{-1}.$$

Whereas $\tilde{\Omega}$ and Ω capture direct input exposures, $\tilde{\Psi}$ and Ψ capture both direct and indirect exposures through the production network. For instance, a higher price of sector h raises sector k's marginal cost directly if sector k uses input h, and indirectly if sector k uses inputs from other sectors that themselves rely on h.

Importantly, the last column of the Leontief inverse measures the fraction of sector i's revenues that is ultimately paid to labor after following all direct and indirect input requirements through the network. Since labor is the only primary input, in the absence of wedges, revenues equal costs at every stage of production, and all costs ultimately trace back to wage payments; that is, $\Psi_{iL} = \tilde{\Psi}_{iL} = 1$ for every sector i. With the wedge above one, however, part of revenues is paid out as operating profits, so revenues exceed wage payments and $\Psi_{iL} < 1$; a sufficiently large subsidy reverses the inequality. Thus, Ψ_{iL} is a measure of the inverse upstream wedge accumulated along sector i's supply chain.

Final expenditure shares and Domar weights. Let $\boldsymbol{b}$ denote the $(N+1) \times 1$ vector of final expenditure shares, whose kth element equals the share of good k in final consumers' expenditure:

$$b_k \equiv \frac{P_k C_k}{\sum_{j=1}^N P_j C_j},$$

with $b_L = 0$ for labor.

The revenue-based Domar weight, or sales share, of sector k is given by

$$\lambda_k = \frac{P_k Y_k}{\sum_{j=1}^N P_j C_j}.$$

In contrast to final expenditure shares, Domar weights measure the exposure of final demand to different sectors both directly and indirectly through the production network.

The last element of $\boldsymbol{\lambda}$, denoted by

$$\Lambda_L = \frac{W\bar{L}}{\sum_{j=1}^{N} P_j C_j},$$

represents the labor income share, or the Domar weight of labor. This object plays a central role in the analysis. Since labor is the primary factor, Λ_L summarizes how much of nominal output accrues to labor after tracing all direct and indirect production requirements through the network. In a frictionless economy with CRS production functions, $\Lambda_L = 1$. Therefore, Λ_L can be interpreted as a measure of the aggregate inverse wedge.

The market-clearing conditions relate sales shares to the Leontief inverse via $\boldsymbol{\lambda}' = \boldsymbol{b}'\Psi$. For comparison, we define the cost-based Domar weights by $\tilde{\boldsymbol{\lambda}}' \equiv \boldsymbol{b}'\tilde{\Psi}$.

Masses of entrants. Since firms within a sector share the same constant-returns-to-scale production technology, operating profits for firm ν in sector k can be written in a symmetric equilibrium as

$$\pi_k(\nu) = \frac{1}{M_k}\frac{1}{\varepsilon_k} P_k Y_k - f_k,$$

where M_k is the mass of firms, ε_k is the perceived price elasticity of demand, and f_k is the fixed overhead cost. The first term is the operating profit earned by each firm, while the second term is the fixed cost of operation.

Free entry implies zero profits, $\pi_k(\nu) = 0$, and therefore determines the equilibrium mass of firms:

$$M_k = \frac{1}{\varepsilon_k}\frac{P_k Y_k}{f_k} = \frac{\lambda_k}{\varepsilon_k \phi_k}. \tag{4}$$

Hence entry rises with the sector's sales share λ_k and the operating margin $1/\varepsilon_k$, and falls with the overhead share ϕ_k. Because the overhead scales with nominal output, the equilibrium mass depends only on the sales *share*, and is therefore invariant to the choice of price numeraire. The mass of firms is not an independent primitive: it is an equilibrium object disciplined by sectoral market size, price elasticities, and overhead costs.

In the main analysis, we assume the overhead is rebated rather than resource-using; thus, the extensive margin is costless from the social planner's perspective. With the boundary conditions $\varepsilon_k(0) = \infty$ and $\eta_k(0) = 0$, the social optimum of the model economy is that within each sector there is a sufficiently large mass of firms $\bar{M}_k$, so that markups converge to one, returns to variety are exhausted, and the economy reaches the perfectly competitive, constant-returns frontier. We focus on the economy with insufficient entry to derive the sufficient statistics in the subsequent section, and leave the case in which the overhead is paid in labor to the Online Appendix D.

3 Sufficient Statistics

This section derives the sufficient statistics for sectoral policy. We first characterize changes in real output through ex post changes in the labor income share, sectoral wedges, and the mass of entrants. We then express the response to subsidies in terms of primitives and decompose it into distortion centrality and entry centrality, corresponding to the intensive and extensive margins. Finally, we characterize the equilibrium entry response and show how sectoral subsidies generate entry cascades across the network.

3.1 Non-Parametric Ex Post Results

We begin with an aggregation theorem that characterizes how real output responds to exogenous shocks in disaggregated economies with input–output linkages, distortions, and an extensive margin of products.

Theorem 1. In response to shocks to productivities and subsidies, the first-order change in real output is:

$$d\log Y = \underbrace{\sum_{k=1}^{N} \tilde{\lambda}_k d\log z_k}_{\Delta \text{ technology}} + \underbrace{\sum_{k=1}^{N} \tilde{\lambda}_k \eta_k d\log M_k}_{\Delta \text{ variety}} \underbrace{- \sum_{k=1}^{N} \tilde{\lambda}_k d\log \mu_k - d\log \Lambda_L}_{\Delta \text{ allocative efficiency}} . \tag{5}$$

Furthermore, following changes in sectoral subsidies, the change in real output can be decomposed into intensive and extensive margins

$$\begin{aligned} d\log Y = & \underbrace{\sum_{k=1}^{N} \left(\tilde{\lambda}_k + \frac{d\log \Lambda_L}{d\log \mu_k} \right) d\log \tau_k}_{\text{intensive margin}} \\ & + \underbrace{\sum_{k=1}^{N} \left[\eta_k \tilde{\lambda}_k + \xi_k \left(\tilde{\lambda}_k + \frac{d\log \Lambda_L}{d\log \mu_k} \right) - \frac{d\log \Lambda_L}{d\log M_k} \right] d\log M_k}_{\text{extensive margin}} . \end{aligned} \tag{6}$$

Theorem 1 provides a non-parametric characterization of the first-order output response in an economy with increasing returns to scale and an extensive margin of products.[2] Specifically, equation (5) decomposes the output response into three components: a technology effect, a variety effect, and a change in allocative efficiency. The first and third components correspond to the aggregation forces in Baqaee and Farhi (2020b). Holding resource allocation and the masses of

[2]Baqaee and Farhi (2020a), *Entry vs. Rents*, develop a general theory of aggregation with entry, exit, and economies of scale in distorted economies, which they use to characterize the social cost of distortions and optimal policy. Theorem 1 covers the same environment but is oriented toward the positive first-order output response to policy that underlies the sufficient statistics developed below.

firms fixed, the first component recovers Hulten's theorem: a productivity change in sector k moves real output in proportion to the sector's importance in forward propagation, $\tilde{\lambda}_k$.[3] The third component, $-\sum_{k=1}^{N} \tilde{\lambda}_k \, d\log \mu_k - d\log \Lambda_L$, captures changes in allocative efficiency. With a single primary factor, reallocation in a distorted economy is summarized by the change in the labor income share, $d\log \Lambda_L$, together with the direct effect of changes in wedges. When there are no wedges, this allocative-efficiency term vanishes.

The new term is the direct variety effect, $\sum_{k=1}^{N} \tilde{\lambda}_k \eta_k \, d\log M_k$. Entry lowers the sectoral price index, $d\log P_k = -\eta_k \, d\log M_k$, so it propagates to real output exactly as a sectoral productivity change does. This ex post decomposition, however, understates the role of entry in the policy experiment: when sectoral subsidies induce entry, the masses of firms affect output not only through the variety effect but also through allocative efficiency.

Equation (6) makes this precise, specializing the output response to changes in sectoral subsidies and separating it into two margins. Holding the masses of firms fixed, the *intensive margin*, in the sense of Liu (2019), captures how a subsidy to sector k changes allocative efficiency: a higher subsidy $d\log \tau_k$ narrows the wedge in sector k, moving output directly through $\tilde{\lambda}_k$ and indirectly through the labor-reallocation term $d\log \Lambda_L / d\log \mu_k$. The *extensive margin* collects the effects of subsidy-induced entry. An increase in the mass of entrants in sector k affects output through the variety effect, $\eta_k \tilde{\lambda}_k$, and through allocative efficiency: a pro-competitive wedge adjustment, $\xi_k\big(\tilde{\lambda}_k + d\log \Lambda_L / d\log \mu_k\big)$, as a more crowded market makes residual demand more elastic, and a reallocation effect, $-d\log \Lambda_L / d\log M_k$.

3.2 Distortion and Entry Centralities

The decomposition into two margins in equation (6) holds for arbitrary technology and consumption functions, but the two margins depend on the equilibrium response of the labor income share Λ_L, which a policymaker cannot observe before acting. This subsection expresses that response through the model's primitives.

To obtain closed-form expressions, we restrict attention to nested-CES production and consumption functions. Specifically, each firm ν in sector k combines labor and intermediate inputs according to

$$y_k(\nu) = z_k \left(\alpha_k^{\frac{1}{\sigma_k}} l_k(\nu)^{\frac{\sigma_k - 1}{\sigma_k}} + \sum_{h=1}^{N} \omega_{kh}^{\frac{1}{\sigma_k}} x_{kh}(\nu)^{\frac{\sigma_k - 1}{\sigma_k}} \right)^{\frac{\sigma_k}{\sigma_k - 1}},$$

where α_k and ω_{kh} are input intensities, and σ_k is the elasticity of substitution across inputs. The final-demand aggregator, in turn, is

$$\mathcal{Y}(\{C_k\}_{k=1}^{N}) = \left(\sum_{k=1}^{N} \beta_k^{\frac{1}{\sigma_0}} C_k^{\frac{\sigma_0 - 1}{\sigma_0}} \right)^{\frac{\sigma_0}{\sigma_0 - 1}},$$

[3] When there are no wedges, cost-based and revenue-based Domar weights coincide, so $\tilde{\lambda}_k = \lambda_k$.

where β_k is the intensity of final expenditure on good k and σ_0 is the elasticity of substitution across sectors.

To simplify the notation, we introduce an input–output covariance operator, following Baqaee and Farhi (2020b). The substitution matrix $\Phi(\boldsymbol{\sigma})$ is an $(N+1)\times(N+1)$ matrix that depends linearly on the cross-sector substitution elasticities $\boldsymbol{\sigma} = \{\sigma_j\}_{j=0}^N$, with (l,k)-th element

$$[\Phi(\boldsymbol{\sigma})]_{l,k} \equiv \sum_{j=0}^{N} \sigma_j \lambda_j \mu_j^{-1} \mathrm{Cov}_{\tilde{\Omega}(j,:)}\left(\frac{\Psi_{(l)}}{\lambda_l}, \frac{\tilde{\Psi}_{(k)}}{\tilde{\lambda}_k}\right)$$
$$= \frac{1}{\lambda_l \tilde{\lambda}_k} \sum_{j=0}^{N} \sigma_j \lambda_j \mu_j^{-1}\left[\sum_{i=1}^{N+1} \tilde{\Omega}_{ji}\Psi_{il}\tilde{\Psi}_{ik} - \left(\sum_{i=1}^{N+1}\tilde{\Omega}_{ji}\Psi_{il}\right)\left(\sum_{i=1}^{N+1}\tilde{\Omega}_{ji}\tilde{\Psi}_{ik}\right)\right]$$

where $\tilde{\Omega}(j,:)$ denotes the jth row of $\tilde{\Omega}$, and $\Psi_{(l)}$ and $\tilde{\Psi}_{(k)}$ the lth and kth columns of Ψ and $\tilde{\Psi}$, respectively. For the household row $j=0$, we use the convention $\tilde{\Omega}(0,:) = \boldsymbol{b}'$, $\lambda_0 = 1$, and $\mu_0 = 1$, so the term with $j=0$ captures substitution in final demand.

The following lemma characterizes how the labor income share responds to changes in ex post wedges and firm entry, expressed in terms of production-network primitives and substitution elasticities.

Lemma 1. The labor income share responds to wedges and the masses of entrants according to

$$\frac{d\log\Lambda_L}{d\log\mu_k} = -\tilde{\lambda}_k - \tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k} \qquad \text{and} \qquad \frac{d\log\Lambda_L}{d\log M_k} = \eta_k\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma}-\mathbf{1})]_{L,k}. \tag{7}$$

Equation (7) reveals that the last row of the substitution matrix is the network kernel governing the response of the labor income share. Some limiting cases are worth mentioning: (i) when there are no wedges, $\Psi_{iL} = 1$ for every sector i and $[\Phi(\boldsymbol{\sigma})]_{L,k} = 0$; (ii) when production and consumption functions are Cobb–Douglas, $[\Phi(\mathbf{1})]_{L,k} = \frac{\lambda_k}{\tilde{\lambda}_k}\frac{\Psi_{kL}}{\Lambda_L} - 1$, while when they are Leontief, $[\Phi(\mathbf{0})]_{L,k} = 0$.

In a heterogeneous-firm economy without input–output linkages (see also the horizontal economy in Section 3.4), this network kernel further reduces to $[\Phi(\mathbf{1})]_{L,k} = \bar{\mu}/\mu_k - 1$, where $\bar{\mu} = [\mathbb{E}_b(\mu^{-1})]^{-1} = 1/\Lambda_L$ is the aggregate wedge. Therefore, we interpret $[\Phi(\mathbf{1})]_{L,k}$ as a measure of relative distortion, with negative (positive) values indicating sectors with above-average (below-average) distortions.

Corollary 1 (Sectoral entry and aggregate distortion)**.** If all cross-sector substitution elasticities are common, $\sigma_j = \sigma$ for $j = 0,\ldots,N$, then

$$\frac{d\log\Lambda_L}{d\log M_k} = (\sigma-1)\,\eta_k\,\tilde{\lambda}_k\,[\Phi(\mathbf{1})]_{L,k},$$

so entry in sector k lowers aggregate distortion if and only if $(\sigma-1)\,[\Phi(\mathbf{1})]_{L,k} > 0$.

The corollary connects the entry margin to the aggregate-market-power question of Edmond et al. (2023). The labor income share is the aggregate inverse wedge, $\Lambda_L = 1/\bar{\mu}$, so entry lowers aggregate market power exactly when it raises Λ_L. Under complementarity ($\sigma < 1$), entry in an above-average-wedge sector draws expenditure away from it toward lower-wedge sectors, lowering the aggregate wedge; under substitutability ($\sigma > 1$) the reallocation reverses, drawing expenditure toward the high-wedge sector and raising it. As in Edmond et al. (2023), whether entry raises or lowers aggregate market power thus depends on the direction of cross-sector reallocation.

Building on Theorem 1 and Lemma 1, Proposition 1 expresses the output response to a subsidy in terms of the model's primitives—the production network, wedges, substitution elasticities, returns to variety, and pro-competitive elasticities. Thus the sufficient-statistic result below is conditional on these objects and on the entry equation implied by free entry; it is not a claim that the entry response can be inferred from input–output shares alone. Throughout the paper, these first-order responses are evaluated at the no-subsidy benchmark $\boldsymbol{\tau} = \mathbf{1}$, where each sector's wedge equals its desired markup, $\mu_k = \varepsilon_k/(\varepsilon_k - 1)$.

Proposition 1. The output response to a sectoral subsidy decomposes into distortion centrality and entry centrality,

$$\frac{d\log Y}{d\log \tau_k} = \underbrace{\frac{\partial \log Y}{\partial \log \tau_k}}_{\text{distortion centrality}} + \underbrace{\sum_{l=1}^{N} \frac{\partial \log Y}{\partial \log M_l}\frac{d\log M_l}{d\log \tau_k}}_{\text{entry centrality}}, \tag{8}$$

where by Lemma 1 the intensive- and extensive-margin responses are

$$\frac{\partial \log Y}{\partial \log \tau_k} = -\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k}, \qquad \frac{\partial \log Y}{\partial \log M_k} = \eta_k \lambda_k \frac{\Psi_{kL}}{\Lambda_L} - (\eta_k + \xi_k)\,\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k}, \tag{9}$$

where $d\log M_l/d\log \tau_k$ is the equilibrium entry response, which Proposition 2 characterizes as a network cascade.

The two centralities are the ex ante counterparts of the two margins in equation (6). Along the intensive margin, *distortion centrality*, $-\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k}$, as in Liu (2019), is a sufficient statistic that summarizes the output response from narrowing sector k's wedge holding the mass of entrants fixed. Unlike the distortion centrality of Liu (2019), which is independent of cross-sector substitution elasticities, ours follows Baqaee and Farhi (2020b) and depends linearly on those elasticities within a nested-CES framework. The difference arises from the treatment of non-tax wedges: Liu (2019) places them only on intermediate inputs and treats them as deadweight losses that are not rebated to households.

The entry-centrality term is new. By aggregating the output response along the extensive margin, *entry centrality* measures how a subsidy to sector k cascades into entry across sectors and thereby affects real output. An increase in the mass of firms in sector l affects output through two channels: the returns to variety η_l, as added varieties lower the sectoral price index, and the pro-competitive effect ξ_l, as a more crowded market compresses wedges and improves allocative efficiency. Entry centrality aggregates these forces across the network. When both are switched off ($\eta_l = \xi_l = 0$), entry carries no first-order output effect and entry centrality is zero.

Distortion centrality and entry centrality are therefore distinct conditional sufficient statistics. The first is a property of the wedges a sector carries; the second is a property of how subsidizing the sector propagates entry through the network, shaped by the returns to variety, the substitution elasticities, the pro-competitive response of markups, and the length of the supply chains involved. A sector with high distortion centrality but weak entry linkages may deliver a muted aggregate output response, whereas a sector with modest distortion centrality but strong entry linkages can generate a large one once the subsidy changes the mass of entrants.

3.3 Firm Entry Cascades

We next relate ex post wedges and the mass of entrants to exogenous subsidy shocks and demonstrate that sectoral subsidies may trigger a cascading response of firm entry across sectors.

First, the wedge $\mu_k = \tau_k^{-1} \frac{\varepsilon_k}{\varepsilon_k - 1}$ responds both to the subsidy and, through the pro-competitive channel, to firm entry:

$$d \log \mu_k = -d \log \tau_k - \xi_k \, d \log M_k. \tag{10}$$

The Dixit–Stiglitz benchmark sets $\xi_k = 0$, so that the wedge declines one-for-one with the subsidy and is independent of entry.

Second, the zero-profit condition (4) ties the mass of firms to the sector's sales share, $M_k = \lambda_k / (\varepsilon_k \phi_k)$. Because the operating margin $1/\varepsilon_k$ now moves with entry through the variable wedge, log-differentiating with the overhead share ϕ_k held fixed yields

$$\Big(1 + (\varepsilon_k - 1)\, \xi_k\Big)\, d \log M_k = d \log \lambda_k, \tag{11}$$

which reduces to $d \log M_k = d \log \lambda_k$ in the Dixit–Stiglitz benchmark $\xi_k = 0$.

Finally, changes in ex post wedges and the mass of entrants induce relative price changes and, in turn, resource reallocations across sectors, reshaping the distribution of sales shares in the network:

$$d \log \boldsymbol{\lambda} = -\underbrace{\Big[\mathrm{diag}(\boldsymbol{\lambda}^{-1}) \Psi' \mathrm{diag}(\boldsymbol{\lambda}) - I\Big] d \log \boldsymbol{\mu}}_{\text{Backward propagation}} + \underbrace{\Phi(\boldsymbol{\sigma} - \mathbf{1}) \mathrm{diag}(\tilde{\boldsymbol{\lambda}}) \Big(\boldsymbol{\eta} \circ d \log \boldsymbol{M} - d \log \boldsymbol{\mu}\Big)}_{\text{Forward propagation}} \tag{12}$$

This relationship also indicates that changes in ex post wedges propagate both backward and forward through the network, whereas—holding wedges fixed—changes in the mass of entrants propagate only forward, reshaping the sales shares (market sizes) of sectors through expenditure substitution.

Taken together, equations (10)–(12) reveal a feedback loop between the mass of entrants and sectoral sales shares, indicating that entry decisions can cascade through the production network. A change in entry in one sector alters marginal costs (via product differentiation and the returns to variety), leading to relative price changes that reallocate resources and reshape the distribution of sales and profits. These shifts, in turn, influence entry decisions across the network.

Proposition 2 consolidates these relationships and characterizes the response of the mass of entrants to subsidies.

Proposition 2 (Entry cascades from sectoral subsidies)**.** The response of the mass of entrants to subsidies is characterized by:

$$\frac{d\log \boldsymbol{M}}{d\log \boldsymbol{\tau}} = \left[I - \Lambda_M + \left(D_\varepsilon - I - \Lambda_\mu\right) D_\xi\right]^{-1} \Lambda_\mu \tag{13}$$

where $\boldsymbol{\xi}$ and $\boldsymbol{\varepsilon}$ are the vectors of pro-competitive elasticities and within-sector substitution elasticities, $D_\xi \equiv \mathrm{diag}(\boldsymbol{\xi})$ and $D_\varepsilon \equiv \mathrm{diag}(\boldsymbol{\varepsilon})$, $\Lambda_\mu = -\frac{\partial \log \boldsymbol{\lambda}}{\partial \log \boldsymbol{\mu}} = \mathrm{diag}(\boldsymbol{\lambda}^{-1})\Psi'\mathrm{diag}(\boldsymbol{\lambda}) - I + \Phi(\boldsymbol{\sigma} - \mathbf{1})\mathrm{diag}(\tilde{\boldsymbol{\lambda}})$, and $\Lambda_M = \frac{\partial \log \boldsymbol{\lambda}}{\partial \log \boldsymbol{M}} = \Phi(\boldsymbol{\sigma} - \mathbf{1})\mathrm{diag}(\tilde{\boldsymbol{\lambda}} \circ \boldsymbol{\eta})$. The Dixit–Stiglitz benchmark $\xi_k = 0$ for all k removes the entire D_ξ term, yielding the entry cascade $(I - \Lambda_M)^{-1} \Lambda_\mu$.

Proposition 2 expresses the equilibrium entry response as a *network multiplier*: the impulse Λ_μ collects the direct effect of the subsidy-induced wedge changes on sales shares, holding the mass of firms fixed, and inverting $I - \Lambda_M + (D_\varepsilon - I - \Lambda_\mu)D_\xi$ propagates it through the network. Two feedbacks operate. The matrix Λ_M is the returns-to-variety feedback: a change in the mass of firms moves prices and reallocates sales shares, just as the Leontief inverse compounds input–output rounds. The terms in D_ξ are the pro-competitive feedback: more entrants compress wedges, which both *dampens* a sector's own entry—the diagonal term $(D_\varepsilon - I)D_\xi$, since a thinner operating margin funds fewer firms—and *propagates* wedge changes across the network through $\Lambda_\mu D_\xi$. Economically, a subsidy sets off a cascade: it lowers a sector's price and wedge and raises its demand for inputs; this reallocates demand—and hence entry—toward the sectors linked to it, which feeds back into sales and entry until the network settles at a new fixed point. The multiplier is a market-structure multiplier: policy changes the number of competitors in one market, and input–output linkages determine how that competitive shock is transmitted to other markets. In the constant-wedge, Cobb–Douglas benchmark ($\boldsymbol{\xi} = \mathbf{0}$ and $\Lambda_M = \mathbf{0}$), the cascade collapses to its first round, $d\log \boldsymbol{M}/d\log \boldsymbol{\tau} = \Lambda_\mu$.

Corollary 2 (Cobb–Douglas with Dixit–Stiglitz demand)**.** When production and consumption are Cobb–Douglas ($\boldsymbol{\sigma} = \mathbf{1}$, so that $\Lambda_M = \mathbf{0}$) and within-sector demand is Dixit–Stiglitz ($\xi_k = 0$ and $\eta_k = 1/(\varepsilon_k - 1)$), Propositions 1 and 2 imply

$$\frac{\partial \log Y}{\partial \log \tau_k} = \tilde{\lambda}_k - \lambda_k \frac{\Psi_{kL}}{\Lambda_L}, \quad \frac{\partial \log Y}{\partial \log M_l} = \frac{\tilde{\lambda}_l}{\varepsilon_l - 1}, \quad \frac{\partial \log M_l}{\partial \log \tau_k} = \frac{\lambda_k}{\lambda_l}(\Psi_{kl} - \delta_{kl}),$$

with $\delta_{kl} = \mathbf{1}(k = l)$, so that

$$\frac{d \log Y}{d \log \tau_k} = \underbrace{\tilde{\lambda}_k - \lambda_k \frac{\Psi_{kL}}{\Lambda_L}}_{\text{distortion centrality}} \underbrace{+ \lambda_k \sum_{l=1}^{N} \frac{\tilde{\lambda}_l}{\varepsilon_l - 1} \frac{\Psi_{kl} - \delta_{kl}}{\lambda_l}}_{\text{entry centrality}}.$$

Cobb–Douglas shuts down expenditure substitution, so sales shares do not respond to entry ($\Lambda_M = \mathbf{0}$), and Dixit–Stiglitz keeps the operating margin constant ($\xi_k = 0$); the entry cascade therefore collapses to its first round. A subsidy to sector k raises its sales share and, through the cost-based linkages, induces entry backward along the supply chain—entry in an upstream sector l whenever $\Psi_{kl} > 0$. Because $\Psi_{kk} \geq 1$, a sector's own entry is weakly positive, $\partial \log M_k / \partial \log \tau_k = \Psi_{kk} - 1 \geq 0$, and each entrant raises output through the variety effect $\tilde{\lambda}_l / (\varepsilon_l - 1)$.

This expression makes the benchmark ranking especially transparent. When cost- and revenue-based Domar weights coincide, $\tilde{\lambda}_l = \lambda_l$, and the within-sector elasticity is common, $\varepsilon_l = \varepsilon$, entry centrality normalized by the sector's sales share is proportional to the row sum of the Leontief inverse, $\sum_{l=1}^{N} \Psi_{kl}$. This row sum is a backward supply-chain multiplier: it measures the total direct and indirect upstream production activated by an expansion of sector k. In this benchmark, high-entry-centrality sectors are therefore sectors whose expansion pulls strongly on their supply chains, not necessarily sectors that are upstream themselves.

Distortion centrality, $\tilde{\lambda}_k - \lambda_k \Psi_{kL} / \Lambda_L$, measures the gap between the cost- and revenue-based Domar weights and vanishes only when wedges are absent. Entry centrality is instead funded by the operating margin $1/\varepsilon_l$—through the variety effect $\tilde{\lambda}_l / (\varepsilon_l - 1)$—so it remains first-order even where distortions are small, which is the sense in which the extensive margin is a genuinely separate channel. It is nonetheless a market-power phenomenon: pushing wedges to unity by sending $\varepsilon_l \to \infty$ (perfect competition) drives both the operating margin $1/\varepsilon_l$ and the returns to variety $\eta_l = 1/(\varepsilon_l - 1)$ to zero, neutralizing entry centrality together with the markup. At any finite ε_l, however, entry centrality is active and can be sizeable even where distortion centrality is small.

More broadly, the two sufficient statistics summarize different features of a sector's position in the production network. Distortion centrality, as in Liu (2019), is largest for *upstream* sectors, whose wedges propagate *forward* to a broad set of downstream users. Entry centrality need not align with this forward-propagation measure: it records the extensive-margin response—how much firm entry a subsidy sets off across the network—and in our calibration is in fact larger for more downstream sectors than for upstream ones. This distinction generates a re-ranking implication: an intensive-margin theory points toward upstream distorted inputs, whereas a market-structure theory can point toward downstream markets whose expansion supports entry throughout their supply chains. Because the two statistics load on different sets of industries, entry centrality conveys information beyond distortion centrality, a complementarity documented in Section 4.

3.4 Illustrative Examples

In this section, we present three simple examples, depicted in Figure 1, to illustrate the intuition behind Proposition 1. All three allow variable markups ($\xi_k \geq 0$): the vertical chain shows how a subsidy sets off an entry cascade upstream along a supply chain; the horizontal economy shuts down intermediate linkages to isolate how a subsidy reallocates entry across sectors through demand; and the roundabout economy collapses the production network to a single sector to show how the intensive and extensive margins interact within it.

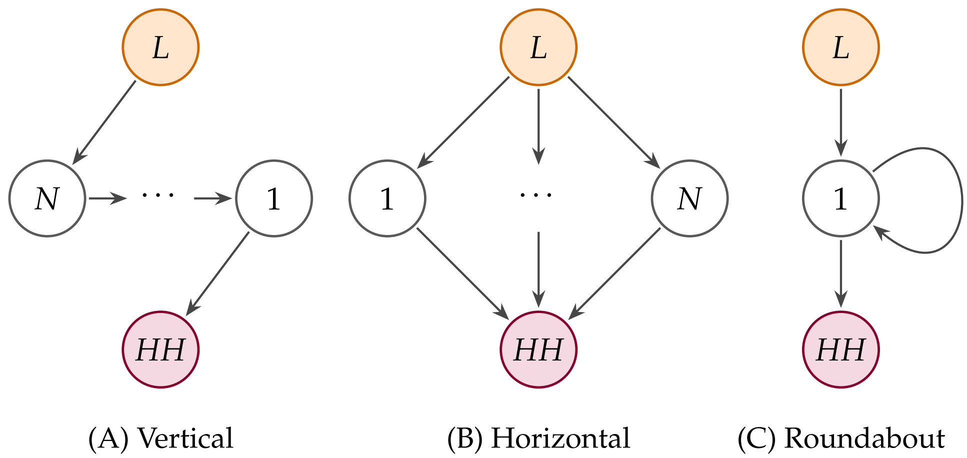

Figure 1: Vertical, Horizontal, and Roundabout Economies. *Notes:* Orange nodes denote labor, purple nodes denote households, and white nodes denote production sectors. Arrows indicate the direction of goods flows.

Example 1 (Vertical Economy)**.** The vertical economy represents a production line where the most upstream sector, N, produces solely using labor. Each downstream sector i processes the output of the sector immediately upstream ($i+1$) linearly, transforming it into its own output. The household consumes the final goods produced by the most downstream sector, sector 1. Propositions 1 and 2 yield

$$\frac{\partial \log Y}{\partial \log \tau_k} = 0, \quad \frac{\partial \log Y}{\partial \log M_j} = \eta_j,$$

and

$$\frac{\partial \log M_j}{\partial \log \tau_k} = \begin{cases} \delta_j \prod_{k<h<j} (1 + \xi_h \delta_h), & j > k, \\ 0, & j \leq k, \end{cases}$$

where $\delta_k = \dfrac{1}{1 + (\varepsilon_k - 1)\xi_k}$.

In this economy, there is only one feasible allocation of resources for a given labor supply, so distortion centrality is trivially zero: holding the mass of entrants fixed, aggregate output is unresponsive to changes in wedges, and a subsidy has no first-order effect along the intensive margin. A subsidy to sector k can nonetheless stimulate firm entry in its upstream sectors, generating an output response along the extensive margin. The induced entry is strictly directional: $\partial \log M_j / \partial \log \tau_k = 0$ for every $j \leq k$, and it is positive only for the upstream sectors $j > k$.

The intuition is simple. Sector 1 is the most downstream sector and sector N is the most upstream sector. Sector k buys from sector $k + 1$, which buys from sector $k + 2$, and so on. A subsidy to sector k lowers sector k's price. Because the chain is linear, this cost reduction is passed downstream to sectors $k-1, k-2, \ldots, 1$, and ultimately to the final good. Quantities for sector k and all downstream sectors are pinned down by the chain technology, so the subsidy does not expand their market size relative to final expenditure. Their nominal sales fall in the same proportion as the final-good price, leaving their sales shares λ_j unchanged for every $j \leq k$. Since free entry ties the mass of firms to the sales share, their entry does not respond. Upstream sectors are different: their prices are not directly reduced by the subsidy, while the final-good price falls. Their sales shares therefore rise relative to final expenditure, raising operating surplus and inducing entry upstream. Variable markups amplify this upstream response: ξ_h in the sectors between k and j intensifies the entry of upstream sector j in response to a subsidy to downstream sector k.

Example 2 (Horizontal Economy)**.** We next consider a case where substitution occurs solely on the demand side: all firms use labor as their only input, while the household combines sectoral outputs through a CES aggregator with elasticity σ_0. We allow variable markups, so the pro-competitive elasticity ξ_k need not vanish. Proposition 1 implies that the distortion centrality is

$$\frac{\partial \log Y}{\partial \log \tau_k} = \sigma_0 b_k \mu_k^{-1}(\mu_k - \bar{\mu}),$$

with $\bar{\mu} = [\mathbb{E}_b(\mu^{-1})]^{-1}$ the expenditure-weighted harmonic mean of wedges. Because the only margin of reallocation is demand-side substitution, this intensive-margin term is unaffected by the pro-competitive channel.

The output response to changes in the mass of entrants is

$$\frac{d \log Y}{d \log M_l} = \eta_l b_l + \Big[(\sigma_0 - 1)\eta_l + \sigma_0 \xi_l\Big] b_l \mu_l^{-1}(\mu_l - \bar{\mu}),$$

and the effect of a subsidy on the mass of entrants is

$$\frac{\partial \log M_l}{\partial \log \tau_k} = (1 - \sigma_0)\, \tilde{\chi}_l \left(b_k \frac{\omega_k}{\mathbb{E}_b(\omega)} - \delta_{kl} \right),$$

where

$$\tilde{\chi}_k = \Big[1 + (1 - \sigma_0)\eta_k + (\varepsilon_k - \sigma_0)\xi_k\Big]^{-1}, \qquad \omega_k = \tilde{\chi}_k \Big[1 + (\varepsilon_k - 1)\xi_k\Big].$$

The pro-competitive elasticity enters twice. Through the $\sigma_0 \xi_l$ term, it raises the output value of each new variety, because entry now also compresses an inefficiently high markup. Through $\tilde{\chi}_k$, it attenuates the entry response itself: the factor shrinks as ξ_k rises, since entry erodes the operating surplus that funds it. Setting $\xi_k = 0$ recovers the Dixit–Stiglitz benchmark, in which $\tilde{\chi}_k = \omega_k = \chi_k \equiv [1 + (1 - \sigma_0)\eta_k]^{-1}$.

In this horizontal economy, the direct effect of a subsidy to sector k on aggregate output linearly depends on the elasticity of substitution in consumption (σ_0) and the sectoral expenditure share (b_k). It is further shaped by the relationship between the sector's wedge (μ_k) and the economy-wide average wedge ($\bar{\mu}$). Subsidizing a sector with an above-average wedge ($\mu_k > \bar{\mu}$) reduces distortions by reallocating resources from low-wedge sectors to high-wedge sectors, thereby enhancing aggregate output. In contrast, subsidizing a sector with a below-average wedge ($\mu_k < \bar{\mu}$) reallocates resources inefficiently to a less-distorted sector, exacerbating misallocation and ultimately reducing output.

The subsidy to sector k also affects output along the extensive margin. Specifically, it reduces sector k's price, causing relative price changes across sectors and driving firm entry through a substitution effect. The direction of resource reallocation depends on the deviation from the Cobb–Douglas case. Under complementarity ($\sigma_0 < 1$), a subsidy to sector k reallocates resources away from sector k to other sectors, encouraging firm entry in those sectors. Conversely, with substitutability ($\sigma_0 > 1$), resources are reallocated toward sector k, leading to firm exits in other sectors.

Corollary 3. In a horizontal economy with homogeneous wedges, the output response to a change in subsidy is

$$\frac{d\log Y}{d\log \tau_k} = \underbrace{0}_{\text{Distortion centrality}} \quad \underbrace{+\,(1-\sigma_0) b_k\, \tilde{\chi}_k\Big[\Big(1 + (\varepsilon_k - 1)\xi_k\Big)\frac{\mathbb{E}_b(\eta\tilde{\chi})}{\mathbb{E}_b(\omega)} - \eta_k\Big]}_{\text{Entry centrality}}.$$

Under Dixit–Stiglitz demand ($\xi_k = 0$) the entry centrality reduces to $(1-\sigma_0)b_k\chi_k[\mathbb{E}_{b\chi}(\eta) - \eta_k]$, with $\mathbb{E}_{b\chi}(\eta) = \mathbb{E}_b(\eta\chi)/\mathbb{E}_b(\chi)$. This is in fact *zero*: with $\eta_k = 1/(\varepsilon_k - 1)$ tied to the wedge, homogeneous wedges make η_k uniform, so $\mathbb{E}_{b\chi}(\eta) = \eta_k$.

In Corollary 3 homogeneous wedges make distortion centrality zero, so the entire output response runs through entry centrality, which measures the output value of the entry that a subsidy to sector k reallocates across the economy. The bracket weighs sector k's own returns to variety, η_k, against a cross-sector benchmark, $(1 + (\varepsilon_k - 1)\xi_k)\,\mathbb{E}_b(\eta\tilde{\chi})/\mathbb{E}_b(\omega)$, so a subsidy reallocates entry between sector k and sectors whose varieties are worth more or less than its own. The prefactor $(1-\sigma_0)\, b_k\, \tilde{\chi}_k$ sets the direction and size of that reallocation through the consumption-substitution sign $(1-\sigma_0)$, the sector's expenditure share b_k, and the attenuation $\tilde{\chi}_k$ that dampens entry as returns to variety and the pro-competitive response rise. Because it is a comparison across sectors, the statistic is nonzero only when sectors differ in η_k or, under variable markups, in ξ_k. Dixit–Stiglitz demand removes both—it ties $\eta_k = 1/(\varepsilon_k - 1)$ to the wedge, uniform under homogeneous wedges, and sets $\xi_k = 0$—so entry centrality is zero as well, and a non-zero extensive margin requires either non-Dixit–Stiglitz heterogeneity in returns to variety or the pro-competitive channel of variable markups.

The sign of this reallocation turns on the consumption elasticity. When sector k's returns to variety fall below the benchmark, a subsidy under complementarity ($\sigma_0 < 1$) reallocates resources toward sectors with higher returns to variety, potentially enhancing aggregate efficiency; under substitutability ($\sigma_0 > 1$) it instead shifts resources toward sector k, leading to inefficiency.

Example 3 (Roundabout Economy)**.** Finally, we consider a roundabout economy with a single sector that combines labor and its own output through a CES production function with non-unitary elasticity ($\sigma_1 \in (0,1)$), together with a general within-sector aggregator with returns to variety η_1 and pro-competitive elasticity $\xi_1 \geq 0$. Collapsing the network to a single sector isolates how the intensive and extensive margins interact once the production structure itself responds to policy. Because the model is linearized at the no-subsidy point $\tau_1 = 1$, the wedge equals the desired markup, $\mu_1 = \varepsilon_1/(\varepsilon_1 - 1)$, so that $1 - \mu_1^{-1} = 1/\varepsilon_1$ is the sector's Lerner index. Propositions 1 and 2 then imply that

$$\frac{\partial \log Y}{\partial \log \tau_1} = \sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)/\varepsilon_1, \quad \frac{\partial \log Y}{\partial \log M_1} = \eta_1 \lambda_1 + (\eta_1 + \xi_1)\, \sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)/\varepsilon_1,$$

and

$$\frac{d \log M_1}{d \log \tau_1} = \frac{\sigma_1(\lambda_1 - 1)}{1 + \eta_1(1-\sigma_1)(\lambda_1 - 1) + \xi_1\left[(\varepsilon_1 - 1) - \sigma_1(\lambda_1 - 1)\right]}.$$

Distortion centrality, $\sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)/\varepsilon_1$, is positive whenever the sector charges a finite markup and uses intermediate inputs ($\varepsilon_1 < \infty$ and $\tilde{\lambda}_1 > 1$): input subsidies then reduce misallocation and raise aggregate output. It scales with the Lerner index $1/\varepsilon_1$, the input substitutability σ_1, and roundaboutness $\tilde{\lambda}_1 - 1$, and, as a pure intensive-margin object, is independent of the pro-competitive elasticity ξ_1 and the returns to variety η_1.

The pro-competitive channel reshapes the extensive margin in two ways. First, it raises the output value of each entrant: beyond the variety effect $\eta_1 \lambda_1$, an entrant also compresses the sector's wedge, adding the allocative-efficiency term $(\eta_1 + \xi_1)\sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)/\varepsilon_1$, so $\partial \log Y / \partial \log M_1$ increases in ξ_1. Second, it changes the entry response: the denominator of $d \log M_1 / d \log \tau_1$ gains the term $\xi_1[(\varepsilon_1 - 1) - \sigma_1(\lambda_1 - 1)]$, which combines a self-dampening force, $(\varepsilon_1 - 1)\xi_1$, since a thinner operating margin funds fewer firms, and a propagation force, $-\sigma_1(\lambda_1 - 1)\xi_1$, operating through the roundabout loop: entry compresses the wedge and, because the sector buys its own output, raises its sales share and fuels further entry. The induced entry is therefore amplified when the roundabout feedback dominates the margin effect, $\sigma_1(\lambda_1 - 1) > \varepsilon_1 - 1$, and dampened otherwise.

Setting $\xi_1 = 0$ recovers the Dixit–Stiglitz case, in which $\eta_1 = 1/(\varepsilon_1 - 1)$ and the entry response reduces to $\sigma_1(\lambda_1 - 1)/[1 + \eta_1(1 - \sigma_1)(\lambda_1 - 1)]$. Because the operating margin is then constant, the subsidy moves entry only through the sales share, $\sigma_1(\lambda_1 - 1)$, which is positive whenever the economy is roundabout ($\lambda_1 > 1$); the entry margin grows with roundaboutness and with the substitution elasticity σ_1, which amplifies both the output value of entry and the sensitivity of entry to the subsidy.

4 Quantitative Analysis

In this section, we use input–output data to quantify distortion centrality and entry centrality across industries. The goal is not to estimate optimal subsidies, but to assess how much the extensive margin changes the output ranking of sectors relative to an intensive-margin benchmark. The quantitative exercise asks whether the extensive margin materially changes the sectors a policymaker would prioritize, or whether it is only a theoretical refinement of distortion centrality.

4.1 Calibration

Our benchmark calibration uses the 2019 U.S. input–output tables from the Bureau of Economic Analysis (BEA). These data discipline the network objects that enter the sufficient statistics: the cost-based input–output matrix, final expenditure shares, revenue-based Domar weights, and sectoral upstreamness. We therefore let the observed production network carry the cross-sector heterogeneity in the benchmark and keep preference, technology, and markup parameters deliberately parsimonious.

For the baseline within-sector aggregator we adopt the standard Dixit–Stiglitz form, $\Upsilon_k(q_k) = q_k^{(\varepsilon_k-1)/\varepsilon_k}$, and set the within-sector elasticity of substitution to $\varepsilon_k = 8$ for all sectors, a value commonly used in the New Keynesian literature. This pins down the desired markup $\varepsilon_k/(\varepsilon_k - 1)$, the returns-to-variety elasticity $\eta_k = 1/(\varepsilon_k - 1)$, and the pro-competitive elasticity $\xi_k = 0$. The benchmark is intentionally conservative for the extensive-margin mechanism: it shuts down markup compression and asks whether entry through returns to variety alone is sufficient to change the policy ranking. Keeping ε_k uniform across sectors also prevents the results from being driven by imposed cross-sector markup heterogeneity; differences between distortion centrality and entry centrality then come from the observed network position of sectors. For the cross-sector elasticities, we set the elasticity of substitution across final consumption goods to $\sigma_0 = 0.9$ and across inputs in production to $\sigma_k = 0.5$ for all sectors, consistent with Atalay (2017).

The robustness exercises then relax the two restrictions that are most relevant for the entry channel. First, we vary the within-sector elasticity ε_k, which changes both the desired markup and the strength of returns to variety. Second, we allow a positive pro-competitive elasticity ξ_k. Rather than treat ξ_k as an unrestricted parameter, we discipline it with firm-level pass-through evidence, following the nonparametric Kimball calibration of Baqaee et al. (2024). Let ρ_k denote desired pass-through: the elasticity of a firm's profit-maximizing price to its marginal cost, holding aggregates fixed. Since entry lowers a producer's relative scale at rate $d\log q_k/d\log M_k = -(1 + \eta_k)$ and the markup moves with scale at $d\log\mu_k/d\log q_k = 1/\rho_k - 1$, the pro-competitive response satisfies $\xi_k = (1/\rho_k - 1)(1 + \eta_k)$. Complete pass-through ($\rho_k = 1$) recovers the Dixit–Stiglitz baseline $\xi_k = 0$, while incomplete pass-through ($\rho_k < 1$) gives $\xi_k > 0$. Estimated pass-throughs run from roughly 0.8 for the largest firms to near unity for the smallest (Amiti et al., 2019), so we vary ξ_k between

0 and 0.3. This range links the variable-markup component of entry centrality to pass-through objects used in modern IO and trade work.

Throughout the quantitative analysis, we report distortion centrality and entry centrality normalized by the revenue-based Domar weight λ_k. Because the fiscal cost of a marginal subsidy scales with the sector's sales, the normalization expresses each statistic as an output response per unit of intervention cost. The reported rankings therefore compare the centrality of subsidies rather than the size of sectors.

4.2 Entry Centrality beyond Distortion Centrality

Figure 2 compares distortion centrality and entry centrality across U.S. industries, normalized by the revenue-based Domar weight. The two measures identify different sectors. Manufacturing and upstream sectors tend to have high distortion centrality, while entry centrality is large for several sectors whose subsidies stimulate entry along broader supply chains. Table 1 reports the sectors with the largest positive and negative entry centrality, and the black circles in Figure 2 mark the five with the highest total centrality.

The main quantitative message is that entry centrality contains substantial information not captured by distortion centrality. The correlation between distortion centrality and entry centrality is −0.398. Moreover, when total centrality—the sum of distortion centrality and entry centrality—is regressed on distortion centrality alone, the residual sum of squares accounts for 37.59% of the total sum of squares. Thus, more than a third of the cross-sector variation in total centrality is left unexplained by the intensive-margin statistic alone. The policy ranking therefore cannot be recovered from Liu-style distortion centrality by adding a monotone adjustment for sector size or upstreamness.

This negative correlation reflects the different network positions summarized by the two statistics. Distortion centrality is closely related to upstreamness: wedges in upstream sectors propagate forward to many downstream users. In the benchmark calibration, distortion centrality has a cross-sector correlation of 0.89 with sectoral upstreamness, whereas entry centrality has a correlation of −0.76, so the two statistics load on opposite ends of the supply chain. Online Appendix Table A.2 reports these correlations together with those against other sectoral measures, for both the United States and China. Upstreamness (**?**) is by far the strongest correlate of either statistic, and its high correlation with distortion centrality confirms Liu (2019); both centralities are otherwise nearly uncorrelated with the revenue-based Domar weight, so the Domar normalization leaves a measure of network position rather than size. Entry centrality instead measures the output value of the entry cascade induced by a subsidy. It depends on how a sector's sales response propagates through the network and on the returns-to-variety value of the induced entry. As a result, sectors with similar distortion centrality can have very different total output responses once entry is incorporated.

Table 1 illustrates how entry centrality re-ranks sectors and how the ranking tracks network position. Housing has low distortion centrality but the highest entry centrality, so most of its total output response comes from the extensive margin; financial vehicles, food and beverage,

Table 1: U.S. Sectors with the Highest and Lowest Entry Centrality

U.S. Industry	Distortion Centrality	Entry Centrality	Total Centrality	Upstreamness
Housing	0.074	0.242	0.316	1.00
Financial vehicles	0.107	0.231	0.338	1.44
Food, beverage, and tobacco	0.172	0.224	0.396	1.96
Motor vehicles and parts	0.193	0.197	0.390	2.40
Water transport	0.104	0.180	0.284	1.93
Forestry and fishing	0.219	−0.061	0.158	4.23
Company management	0.166	−0.024	0.142	3.78
Computers and electronics	0.077	−0.014	0.063	3.24
Administrative and support services	0.175	−0.011	0.164	3.77
Primary metals	0.356	−0.001	0.354	4.70

Notes: Top panel reports the five U.S. sectors with the highest entry centrality; bottom panel the five with the lowest. Distortion, entry, and total centrality are normalized by the revenue-based Domar weight; upstreamness is the output-upstreamness measure (distance to final use). All quantities use the 2019 BEA summary input–output table under the benchmark calibration.

and motor vehicles also carry large entry components. These sectors sit close to final demand, with upstreamness near one to two. By contrast, the sectors with the lowest entry centrality—forestry and fishing, company management, computers and electronics, and primary metals—are far upstream, with upstreamness above three, so a subsidy triggers little net entry and their total centrality is close to their distortion centrality alone. An intensive-margin criterion would therefore misrank the output effects of sectoral subsidies. The empirical counterpart of the theory is that downstream markets can be central for industrial policy not because they supply many sectors directly, but because expanding them supports entry and competitive responses in the markets that supply them.

4.3 Reduced-Form Evidence from China

We next examine whether the two centrality measures are related to observed sectoral interventions in China. This exercise is descriptive. It does not estimate the causal effect of centrality on policy choices, nor does it imply that policymakers explicitly compute the sufficient statistics derived above. Instead, it asks whether sectors that receive more favorable policy treatment are systematically related to the intensive- and extensive-margin network statistics highlighted by the model. The purpose is external discipline, not structural validation: if entry centrality is a meaningful policy statistic, it should help organize real-world intervention patterns beyond the existing distortion-centrality measure. We focus on China because the exercise requires sector-level measures of both policy targeting and the support that targeted sectors receive—provided here by China's industrial plans and firm-level data on credit and fiscal treatment—whereas comparable sectoral measures are not readily available for the more decentralized U.S. setting, which we use instead for calibration.

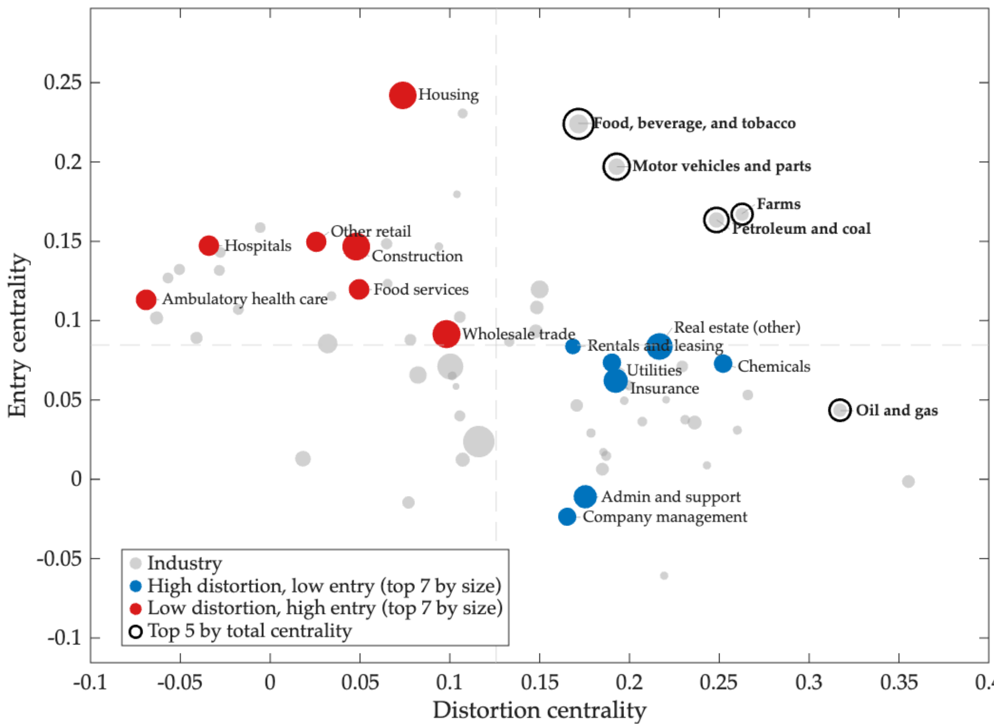

Figure 2: Distortion Centrality and Entry Centrality across U.S. Industries. *Notes:* Each marker is one of the 66 industries in the 2019 BEA summary input–output table, located by its distortion centrality on the horizontal axis and entry centrality on the vertical axis, both normalized by the revenue-based Domar weight. Marker area is proportional to the sector's Domar weight. Dashed lines mark cross-sector means. Blue markers denote the seven largest sectors with above-average distortion centrality and below-average entry centrality; red markers denote the seven largest sectors with above-average entry centrality and below-average distortion centrality. Black circles mark the five sectors with the highest total centrality (distortion plus entry centrality), with their labels set in bold.

We recalibrate the cost-based input–output matrix using China's 2007 input–output table, disaggregated at the 135 three-digit-sector level. We compute distortion centrality and entry centrality under the benchmark calibration, in which the within-sector elasticity $\varepsilon_k = 8$ pins down the desired markup $\varepsilon_k/(\varepsilon_k - 1)$ in every sector, as in the U.S. analysis.[4]

Across China's 135 sectors, the two centralities are negatively related, just as in the United States: sectors with high distortion centrality tend to have low entry centrality, and vice versa. The divergence is also economically systematic—the large high-distortion, low-entry sectors are upstream energy and heavy-materials industries (electric power, petroleum, coal mining, basic chemicals, steel), while the large high-entry, low-distortion sectors are downstream services and primary activities (public administration, education, health care, real estate, and trade). A ranking based only on distortion centrality therefore points to a different set of sectors than one that also accounts for entry centrality; Online Appendix Figure A.1 plots the full cross-section.

To connect these statistics to observed policy, we use the sectoral intervention measures of Liu (2019), built from the 2007 Chinese Annual Survey of Industrial Firms: the effective interest rates and debt-to-capital ratios of private firms, and the fraction of firms receiving tax incentives. These outcomes proxy the more favorable financing and fiscal treatment that targeted sectors receive. The sample covers 79 three-digit manufacturing sectors—the finest partition that can be matched consistently between the national input–output table and the firm-level data—and connects to recent work on Chinese industrial policy and sectoral credit allocation (Chen et al., 2026).

We estimate the following cross-sector regression:

$$\text{Outcome}_i = \alpha + \beta\,\text{Distortion centrality}_i + \gamma\,\text{Entry centrality}_i + X_i'\delta + \epsilon_i,$$

where each observation is a sector and regressions are weighted by sectoral value added. The control vector X_i follows Liu (2019) and includes capital intensity, the Lerner index, average log fixed assets of firms in their first year of operation as a proxy for minimum operating scale, and export intensity.

Table 2 reports the results. The signs of the coefficients are consistent with more favorable treatment for sectors with higher centrality. Higher distortion centrality and higher entry centrality are associated with lower effective interest rates and higher debt ratios. Entry centrality also predicts tax incentives in specifications without controls, although the relationship is weaker once controls are included. The strongest pattern is for effective interest rates: adding entry centrality raises the R^2 from 0.104 to 0.485 without controls and from 0.343 to 0.556 with controls. For debt ratios, adding entry centrality raises the R^2 from 0.214 to 0.355 without controls and from 0.262 to 0.371 with controls.

These results should be interpreted as reduced-form correlations. They do not establish that entry centrality causally determines policy allocation. They do show, however, that the extensive-margin statistic contains explanatory power for observed sectoral interventions beyond distortion

[4] As shown in Section 4.4, both distortion centrality and entry centrality are highly correlated across alternative calibrations of the desired markup.

Table 2: The Role of Entry Centrality in Explaining Sectoral Interventions in China

	Effective Interest Rate				Debt Ratio				Tax Break			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Distortion cent.	-2.037***	-10.192***	-3.022***	-9.709***	7.765***	20.945***	8.000***	20.757***	6.814*	31.149***	11.890***	23.790**
	(0.681)	(1.207)	(0.689)	(1.274)	(1.695)	(3.588)	(1.943)	(4.029)	(3.928)	(8.642)	(4.259)	(9.440)
Entry cent.		-16.455***		-14.577***		26.593***		27.807***		49.099***		25.941
		(2.197)		(2.482)		(6.534)		(7.851)		(15.736)		(18.395)
Capital intensity			-0.424**	-0.379**			-0.433	-0.519			1.089	1.009
			(0.200)	(0.166)			(0.564)	(0.524)			(1.235)	(1.228)
Lerner index			0.117	0.044			-0.215	-0.076			-1.229	-1.100
			(0.181)	(0.150)			(0.509)	(0.475)			(1.116)	(1.112)
Log (fixed assets)			0.045	-0.074			0.349	0.575			-1.085	-0.873
			(0.209)	(0.175)			(0.590)	(0.552)			(1.294)	(1.294)
Export intensity			-0.873***	-0.400**			0.797	-0.106			3.508***	2.666**
			(0.176)	(0.167)			(0.497)	(0.528)			(1.090)	(1.237)
Constant	5.514***	12.510***	6.037***	11.918***	50.512***	39.206***	50.310***	39.090***	27.022***	6.147	24.039***	13.572*
	(0.413)	(0.986)	(0.416)	(1.059)	(1.029)	(2.932)	(1.173)	(3.350)	(2.386)	(7.062)	(2.571)	(7.849)
Observations	79	79	79	79	79	79	79	79	79	79	79	79
R-squared	0.104	0.485	0.343	0.556	0.214	0.355	0.262	0.371	0.038	0.147	0.192	0.213

Notes: Within each outcome block, even-numbered columns add entry centrality, and columns (3)–(4), (7)–(8), and (11)–(12) add the Liu (2019) controls. Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

centrality and standard sectoral controls. The evidence is therefore consistent with the paper's main mechanism: industrial policies are related not only to pre-existing distortions along the intensive margin, but also to sectors whose subsidies are associated with larger entry spillovers through the production network. The empirical claim is deliberately modest: the new statistic is not redundant in the policy environment that motivated the original distortion-centrality approach.

4.4 Robustness

The preceding subsections use the benchmark calibration to show that entry centrality changes sectoral rankings and then relate the two statistics to observed interventions in China. A natural concern is that the quantitative ranking may depend on the elasticities used to map the input–output tables into centrality measures. This subsection addresses that concern by varying the main parameters that govern substitution, returns to variety, and the pro-competitive response of markups, and by comparing the resulting rankings with the benchmark.

Figure 3 reports, for both distortion centrality and entry centrality, the correlation between the statistic under an alternative parameterization and its value under the benchmark calibration. Each of the four panels varies one parameter while holding the others at their benchmark values, so every series equals one at the benchmark (the dotted line). Panels (a) and (b) vary the cross-sector elasticities of substitution in consumption (σ_0) and in production (σ_k); both centralities respond, with the correlation falling as substitution patterns move away from the benchmark and the cross-sector covariance structure of the network changes. Panel (c) varies the within-sector elasticity ε_k, equivalently the desired markup $\varepsilon_k/(\varepsilon_k - 1)$; because ε_k is varied uniformly across sectors the implied markups move little, and both correlations stay close to one. Panel (d) varies the pro-competitive elasticity ξ_k: distortion centrality is invariant to ξ_k—which acts only on the extensive margin—so its correlation is identically one, while entry centrality's correlation declines as ξ_k rises and entry feeds back into markups along the cascade.

These exercises reinforce the distinction between the two sufficient statistics. Both load on the cross-sector substitution elasticities σ_0 and σ_k that shape the covariance structure of the network, so both correlations fall in panels (a) and (b). They diverge on the pro-competitive elasticity: only entry centrality responds to ξ_k (panel d), because ξ_k governs how entry feeds back into markups and so operates purely through the extensive margin, leaving distortion centrality unchanged. The within-sector elasticity ε_k enters both—through the desired markup—but only mildly when it is varied proportionally across sectors.

Additional parameterizations for the motor-vehicles sector show that distortion centrality varies approximately linearly with the cross-sector elasticity of substitution, while entry centrality is more convex in this elasticity. They also show that pro-competitive variable markups in other sectors can amplify entry centrality by strengthening the response of upstream entry to downstream subsidies.

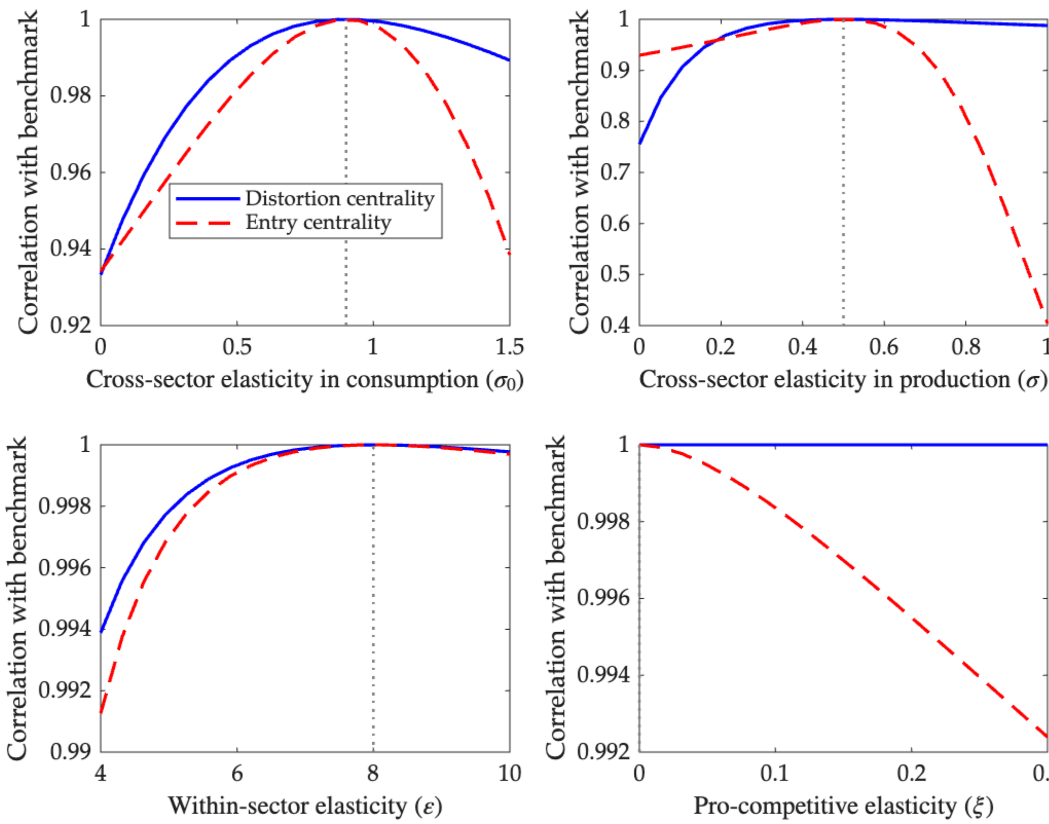

Figure 3: Distortion and Entry Centrality across Specifications. *Notes:* Each panel reports the correlation between a centrality under an alternative parameterization and its value under the benchmark calibration, for distortion centrality (blue, solid) and entry centrality (red, dashed). In reading order, the panels vary the cross-sector elasticity of substitution in consumption (σ_0), the cross-sector elasticity in production (σ), the within-sector elasticity (ε), and the pro-competitive elasticity (ξ); all other parameters are held at the benchmark. The dotted line marks the benchmark value, at which each correlation equals one. Both centralities are normalized by the revenue-based Domar weight.

5 Conclusion

This paper develops a sufficient-statistic framework for the output effects of sectoral subsidies in production networks with endogenous firm entry. Industrial policy changes market structure, not only wedges: by changing sectoral sales, it changes the mass of firms, the set of varieties, and, under variable markups, competitive pressure. The output response therefore combines distortion centrality, the intensive-margin gain from reducing wedges, with entry centrality, the output value of the induced entry cascade.

The two statistics are conceptually and quantitatively distinct. In the U.S. calibration, entry centrality reorders sectors relative to an intensive-margin benchmark, shifting attention toward sectors whose subsidies mobilize entry along their supply chains. In reduced-form evidence from China, entry centrality is also related to observed sectoral interventions and adds explanatory power beyond distortion centrality. The analysis focuses on aggregate output rather than full welfare, since free entry may be excessive or insufficient once business stealing, innovation, or other dynamic forces are introduced. The main lesson is nevertheless simple: evaluating sectoral policy requires combining where distortions are located with how entry and competitive pressure propagate through the network.

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Online Appendix

A Additional Figures and Tables

Figure A.1 plots distortion centrality and entry centrality across China's 135 three-digit sectors, the cross-section summarized in Section 4.

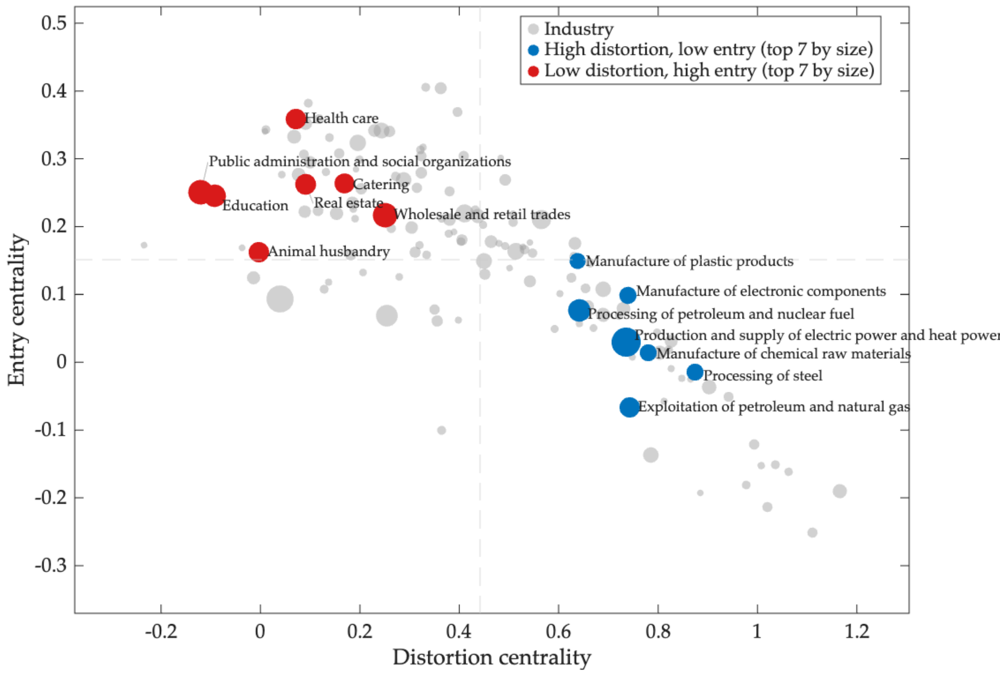

Figure A.1: Distortion and Entry Centrality across Chinese Industries. *Notes:* Each marker is one of China's 135 three-digit sectors, located by its distortion centrality on the horizontal axis and entry centrality on the vertical axis. Marker area is proportional to the sector's Domar weight. Dashed lines mark cross-sector means. Blue markers denote the seven largest sectors with above-average distortion centrality and below-average entry centrality; red markers denote the seven largest sectors with above-average entry centrality and below-average distortion centrality. Both centralities are computed from China's 2007 input–output table with benchmark calibration ($\varepsilon = 8$, $\sigma_0 = 0.9$, $\sigma = 0.5$, $\xi = 0$).

Table A.1 reports distortion centrality, entry centrality, total centrality, and upstreamness for all 66 U.S. industries under the benchmark calibration, sorted by total centrality. It is the full version of the rankings summarized in Table 1: entry centrality is concentrated in downstream, low-upstreamness sectors and fades as upstreamness rises.

Table A.1: Distortion Centrality, Entry Centrality, and Upstreamness

Industry	Sales Share	Distortion	Entry	Total	Upstreamness
Farms	0.022	+0.263	+0.167	+0.430	3.04
Petroleum and coal	0.032	+0.249	+0.163	+0.412	3.10
Food, beverage, and tobacco	0.053	+0.172	+0.224	+0.396	1.96
Motor vehicles and parts	0.038	+0.193	+0.197	+0.390	2.40
Oil and gas	0.023	+0.317	+0.043	+0.361	4.26
Primary metals	0.020	+0.356	−0.001	+0.354	4.70
Financial vehicles	0.009	+0.107	+0.231	+0.338	1.44
Chemicals	0.050	+0.252	+0.073	+0.325	3.76
Paper products	0.012	+0.266	+0.053	+0.319	3.91
Housing	0.119	+0.074	+0.242	+0.316	1.00
Plastics and rubber	0.016	+0.229	+0.071	+0.301	3.54
Real estate (other)	0.111	+0.217	+0.084	+0.300	3.40
Mining (non-oil/gas)	0.007	+0.260	+0.031	+0.291	4.05
Water transport	0.003	+0.104	+0.180	+0.284	1.93
Fabricated metals	0.025	+0.236	+0.035	+0.272	3.81
Textiles and mills	0.003	+0.221	+0.050	+0.271	3.62
Broadcasting	0.046	+0.150	+0.120	+0.270	2.75
Wood products	0.008	+0.231	+0.037	+0.268	3.77
Utilities	0.049	+0.190	+0.073	+0.263	3.37
Nonmetal minerals	0.010	+0.200	+0.059	+0.259	3.41
Machinery	0.023	+0.149	+0.108	+0.257	2.84
Insurance	0.091	+0.192	+0.062	+0.254	3.44
Rentals and leasing	0.034	+0.168	+0.084	+0.252	3.15
Pipeline transport	0.004	+0.243	+0.009	+0.252	4.11
Rail transport	0.005	+0.197	+0.050	+0.247	3.58
Electrical equipment	0.009	+0.207	+0.036	+0.243	3.62
Truck transport	0.024	+0.148	+0.094	+0.242	2.94
Furniture	0.005	+0.094	+0.147	+0.241	2.01
Movies and recording	0.011	+0.133	+0.087	+0.219	2.89
Data processing and IT services	0.020	+0.171	+0.046	+0.217	3.42
Other transportation equipment	0.014	+0.065	+0.149	+0.213	1.82
Waste management	0.007	+0.179	+0.029	+0.208	3.58
Air transport	0.016	+0.105	+0.102	+0.208	2.58
Printing	0.004	+0.186	+0.017	+0.203	3.65
Warehousing	0.010	+0.187	+0.015	+0.202	3.65
Construction	0.121	+0.048	+0.147	+0.195	1.70
Other transport and support	0.020	+0.185	+0.006	+0.191	3.74
Wholesale trade	0.123	+0.098	+0.091	+0.190	2.63
Mining support	0.009	+0.065	+0.123	+0.188	2.19
Other retail	0.062	+0.025	+0.150	+0.175	1.50
Banking and credit	0.105	+0.100	+0.071	+0.171	2.69

continued on next page

Industry	Sales Share	Distortion	Entry	Total	Upstreamness
Food services	0.064	+0.049	+0.120	+0.169	2.01
Transit and passenger transport	0.006	+0.101	+0.065	+0.166	2.85
Accommodation	0.016	+0.078	+0.088	+0.166	2.52
Admin and support	0.083	+0.175	−0.011	+0.164	3.77
Apparel and leather	0.001	+0.103	+0.058	+0.162	2.88
Forestry and fishing	0.004	+0.219	−0.061	+0.158	4.23
Amusement and recreation	0.012	−0.006	+0.159	+0.153	1.14
Misc. manufacturing	0.008	+0.034	+0.115	+0.150	1.88
Securities and investments	0.043	+0.082	+0.066	+0.148	2.67
Performing arts and museums	0.013	+0.106	+0.040	+0.145	3.06
Company management	0.048	+0.166	−0.024	+0.142	3.78
Professional services	0.157	+0.116	+0.024	+0.140	3.22
Legal services	0.026	+0.107	+0.013	+0.120	3.26
Other services (non-gov)	0.056	+0.032	+0.086	+0.118	2.19
Food and beverage stores	0.013	−0.028	+0.143	+0.116	1.10
Hospitals	0.063	−0.034	+0.147	+0.113	1.00
General merchandise	0.013	−0.029	+0.132	+0.103	1.23
Motor vehicle dealers	0.013	−0.018	+0.107	+0.089	1.57
Nursing and residential care	0.014	−0.050	+0.133	+0.082	1.00
Social assistance	0.013	−0.057	+0.127	+0.070	1.00
Computers and electronics	0.019	+0.077	−0.014	+0.063	3.24
Publishing (excl. internet)	0.018	−0.041	+0.089	+0.048	1.57
Ambulatory health care	0.064	−0.069	+0.113	+0.044	1.04
Educational services	0.022	−0.063	+0.102	+0.038	1.21
Computer systems design	0.034	+0.018	+0.013	+0.031	2.70

Notes: All 66 industries in the 2019 BEA summary input–output table, sorted by total centrality from high to low. Distortion, entry, and total centrality are normalized by the sales share; upstreamness is the measure of distance to final use. Benchmark calibration ($\varepsilon = 8$, $\sigma_0 = 0.9$, $\sigma = 0.5$, $\xi = 0$).

Table A.2 reports, for both the United States and China, the cross-sector correlation of distortion centrality and entry centrality with the five sectoral measures discussed in Section 4.

Table A.2: Distortion and Entry Centrality Correlate with Upstreamness, Not Sectoral Size

	United States		China	
	Distortion	Entry	Distortion	Entry
Distortion vs. entry centrality	−0.40		−0.79	
Upstreamness	+0.89	−0.76	+0.96	−0.87
Sales share	−0.09	+0.14	−0.13	+0.02
Consumption share	−0.42	+0.47	−0.45	+0.30
Sectoral value added	−0.28	−0.09	−0.31	+0.03
Intermediate-input expenditure share	+0.63	+0.42	+0.52	+0.02

Notes: Cross-sector Pearson correlations under the benchmark calibration; both centralities are normalized by the sales share. The last row reports the correlation between the two centralities.

B Related Proofs

This appendix collects the proofs of the results stated in the main text.

Proof of Theorem 1. We first derive the price response and then use the labor-income identity to obtain the output response. By Shephard's lemma, the change in sector k's marginal cost is

$$d\log mc_k = -d\log z_k + \sum_{h=1}^{N} \tilde{\Omega}_{kh} d\log P_h + \tilde{\Omega}_{kL} d\log W \tag{B-1}$$

where $\tilde{\Omega}_{kh} = P_h x_{kh}(\nu)/(mc_k y_k(\nu))$ is the cost share of input h in sector k.

The sectoral price index inherits the returns to variety: holding firm-level prices fixed, it declines with the mass of firms at elasticity η_k, so that

$$\begin{aligned} d\log P_k &= -\eta_k d\log M_k + d\log p_k(\nu) \\ &= -\eta_k d\log M_k + d\log \mu_k + d\log mc_k \\ &= -\eta_k d\log M_k + d\log \mu_k - d\log z_k + \sum_{h=1}^{N} \tilde{\Omega}_{kh} d\log P_h + \tilde{\Omega}_{kL} d\log W. \end{aligned} \tag{B-2}$$

Stacking sectors and premultiplying by the cost-based Leontief inverse gives

$$d\log \boldsymbol{P} = \sum_{k=1}^{N} \tilde{\Psi}_{(k)} \left[-\eta_k d\log M_k - d\log z_k + d\log \mu_k\right] + d\log W\,\mathbf{1}. \tag{B-3}$$

Therefore the change in the final-use price index is

$$d\log P^Y = \boldsymbol{b}' d\log \boldsymbol{P} = \sum_{k=1}^{N} \tilde{\lambda}_k \left[-\eta_k d\log M_k - d\log z_k + d\log \mu_k\right] + d\log W. \tag{B-4}$$

Because labor is the only primary factor and is supplied inelastically, nominal labor income satisfies $W\bar{L} = \Lambda_L P^Y Y$. Log-differentiating gives

$$d\log W = d\log \Lambda_L + d\log P^Y + d\log Y. \tag{B-5}$$

Combining the previous two equations yields

$$d\log Y = d\log W - d\log P^Y - d\log \Lambda_L \tag{B-6}$$

$$= \sum_{k=1}^{N} \tilde{\lambda}_k \left(d\log z_k + \eta_k d\log M_k - d\log \mu_k\right) - d\log \Lambda_L, \tag{B-7}$$

which proves equation (5). For the subsidy decomposition, set $d\log z_k = 0$ and expand

$$d\log\Lambda_L = \sum_k \frac{d\log\Lambda_L}{d\log\mu_k}\, d\log\mu_k + \sum_k \frac{d\log\Lambda_L}{d\log M_k}\, d\log M_k.$$

Substituting the wedge relation (10), $d\log\mu_k = -d\log\tau_k - \xi_k\, d\log M_k$, and collecting separately the $d\log\tau_k$ and $d\log M_k$ terms gives equation (6). □

Proof of Lemma 1. The proof has three steps. First, we compute how the revenue-based input–output matrix changes. Since

$$\Omega_{ji} = \mu_j^{-1}\tilde{\Omega}_{ji} = \frac{P_i x_{ji}(\nu)}{\mu_j \sum_h P_h x_{jh}(\nu)},$$

CES input demand implies

$$\begin{aligned} d\Omega_{ji} &= \Omega_{ji}\left[-d\log\mu_j + (1-\sigma_j)\left(d\log P_i - \sum_l \tilde{\Omega}_{jl} d\log P_l\right)\right] \\ &= -\Omega_{ji} d\log\mu_j + (1-\sigma_j)\mu_j^{-1}\mathrm{Cov}_{\tilde{\Omega}(j,:)}(d\log \boldsymbol{P}, I_{(i)}). \end{aligned} \tag{B-8}$$

Second, we translate this into a change in the Leontief inverse. Since $d\Psi = \Psi(d\Omega)\Psi$,

$$\begin{aligned} d\Psi_{mn} &= \sum_{j=1}^{N}\sum_i \Psi_{mj} d\Omega_{ji}\Psi_{in} \\ &= -\sum_{j=1}^{N}\Psi_{mj} d\log\mu_j \sum_i \Omega_{ji}\Psi_{in} + \sum_{j=1}^{N}\Psi_{mj}\mu_j^{-1}(1-\sigma_j)\mathrm{Cov}_{\tilde{\Omega}(j,:)}(d\log \boldsymbol{P}, \Psi_{(n)}). \end{aligned} \tag{B-9}$$

Using $\Omega\Psi = \Psi - I$, this becomes

$$d\Psi_{mn} = -\sum_{j=1}^{N}\Psi_{mj}(\Psi_{jn} - \delta_{jn})\, d\log\mu_j + \sum_{j=1}^{N}\Psi_{mj}\mu_j^{-1}(1-\sigma_j)\mathrm{Cov}_{\tilde{\Omega}(j,:)}(d\log \boldsymbol{P}, \Psi_{(n)}), \tag{B-10}$$

where δ_{jn} is the jn-th element of the identity matrix. Premultiplying by final expenditure shares gives the production-network part of the sales-share response. The final-demand shares also move with relative prices; under CES demand, that contribution is the same covariance term with the household row $j = 0$. Combining the two gives

$$\begin{aligned} \lambda_l d\log\lambda_l = &-\sum_{j=1}^{N}\lambda_j(\Psi_{jl} - \delta_{jl})\, d\log\mu_j \\ &+ \sum_{j=0}^{N}(1-\sigma_j)\lambda_j\mu_j^{-1}\mathrm{Cov}_{\tilde{\Omega}(j,:)}(d\log \boldsymbol{P}, \Psi_{(l)}). \end{aligned} \tag{B-11}$$

Setting $l = L$ and normalizing by $\Lambda_L = \lambda_L$ gives

$$d\log\Lambda_L = -\sum_{j=1}^{N}\lambda_j\frac{\Psi_{jL}}{\Lambda_L}\,d\log\mu_j + \sum_{j=0}^{N}(1-\sigma_j)\lambda_j\mu_j^{-1}\mathrm{Cov}_{\tilde{\Omega}(j,:)}\left(d\log\boldsymbol{P},\frac{\Psi_{(L)}}{\Lambda_L}\right). \tag{B-12}$$

Third, substitute the sectoral-price relation (B-3) into the covariance in (B-12). A productivity shock enters the covariance with coefficient $-\tilde{\Psi}_{(k)}$, a wedge shock with coefficient $\tilde{\Psi}_{(k)}$, and an entry shock with coefficient $-\eta_k\tilde{\Psi}_{(k)}$. The definition of Φ collects these covariance terms, yielding

$$d\log\Lambda_L = \sum_{k=1}^{N}\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma}-\mathbf{1})]_{L,k}\,d\log z_k - \sum_{k=1}^{N}\tilde{\lambda}_k\left(1+[\Phi(\boldsymbol{\sigma})]_{L,k}\right)d\log\mu_k + \sum_{k=1}^{N}\eta_k\,\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma}-\mathbf{1})]_{L,k}\,d\log M_k. \tag{B-13}$$

Reading off the coefficients on $d\log\mu_k$ and $d\log M_k$ proves the two derivatives in Lemma 1.

Finally, we verify the Cobb–Douglas limiting case stated after the lemma. Setting $\boldsymbol{\sigma} = \mathbf{1}$ in the definition of Φ and using $\lambda_L = \Lambda_L$,

$$[\Phi(\mathbf{1})]_{L,k} = \frac{1}{\Lambda_L\tilde{\lambda}_k}\sum_{j=0}^{N}\frac{\lambda_j}{\mu_j}\left[\sum_i\tilde{\Omega}_{ji}\Psi_{iL}\tilde{\Psi}_{ik} - \Big(\sum_i\tilde{\Omega}_{ji}\Psi_{iL}\Big)\Big(\sum_i\tilde{\Omega}_{ji}\tilde{\Psi}_{ik}\Big)\right].$$

Three accounting identities collapse the bracket. First, the revenue-Domar balance

$$\sum_{j=0}^{N}\lambda_j\mu_j^{-1}\tilde{\Omega}_{ji} = \lambda_i$$

follows from $\boldsymbol{\lambda}' = \boldsymbol{b}'\Psi$, that is, from $\lambda_i = b_i + \sum_{j\geq 1}\lambda_j\Omega_{ji}$ with $\Omega_{ji} = \mu_j^{-1}\tilde{\Omega}_{ji}$ and the household convention $b_i = \tilde{\Omega}_{0i}$. It reduces the leading term to $\sum_i\lambda_i\Psi_{iL}\tilde{\Psi}_{ik}$, in which the labor row $i = L$ drops out because $\tilde{\Psi}_{Lk} = 0$.

Second, $\Omega\Psi = \Psi - I$ gives

$$\sum_i\tilde{\Omega}_{ji}\Psi_{iL} = \mu_j(\Psi_{jL} - \delta_{jL}) = \mu_j\Psi_{jL},$$

with the household row $j = 0$ contributing $\sum_i b_i\Psi_{iL} = \Lambda_L$. Third, $\tilde{\Omega}\tilde{\Psi} = \tilde{\Psi} - I$ gives $\sum_i\tilde{\Omega}_{ji}\tilde{\Psi}_{ik} = \tilde{\Psi}_{jk} - \delta_{jk}$, with $j = 0$ contributing $\tilde{\lambda}_k$.

Multiplying the last two identities and summing over j, the product term equals

$$\Lambda_L \tilde{\lambda}_k + \sum_{j=1}^{N} \lambda_j \Psi_{jL} \tilde{\Psi}_{jk} - \lambda_k \Psi_{kL}.$$

The interior sum is term-by-term identical to the leading term, so the two cancel and only $\lambda_k \Psi_{kL} - \Lambda_L \tilde{\lambda}_k$ survives. What remains is

$$[\Phi(\mathbf{1})]_{L,k} = \frac{\lambda_k \Psi_{kL} - \Lambda_L \tilde{\lambda}_k}{\Lambda_L \tilde{\lambda}_k} = \frac{\lambda_k}{\tilde{\lambda}_k} \frac{\Psi_{kL}}{\Lambda_L} - 1.$$

□

Proof of Proposition 1. Set $d \log z_k = 0$ in Theorem 1. Substituting the labor-share response from Lemma 1 gives

$$\begin{aligned} d \log Y = & \sum_{k=1}^{N} \tilde{\lambda}_k \eta_k \, d \log M_k - \sum_{k=1}^{N} \tilde{\lambda}_k \, d \log \mu_k \\ & + \sum_{k=1}^{N} \left(\tilde{\lambda}_k + \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma})]_{L,k} \right) d \log \mu_k - \sum_{k=1}^{N} \eta_k \, \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma} - \mathbf{1})]_{L,k} \, d \log M_k. \end{aligned} \tag{B-14}$$

The wedge terms reduce to

$$\sum_k \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma})]_{L,k} \, d \log \mu_k.$$

For the entry terms, use the linearity of Φ in the elasticity vector:

$$\Phi(\boldsymbol{\sigma} - \mathbf{1}) = \Phi(\boldsymbol{\sigma}) - \Phi(\mathbf{1}).$$

The Cobb–Douglas limiting case in Lemma 1 gives $[\Phi(\mathbf{1})]_{L,k} = \frac{\lambda_k}{\tilde{\lambda}_k} \frac{\Psi_{kL}}{\Lambda_L} - 1$. Thus the coefficient on $d \log M_k$ is $\eta_k \left(\lambda_k \frac{\Psi_{kL}}{\Lambda_L} - \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma})]_{L,k} \right)$, and therefore

$$\begin{aligned} d \log Y = & \sum_{k=1}^{N} \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma})]_{L,k} \, d \log \mu_k \\ & + \sum_{k=1}^{N} \eta_k \left(\lambda_k \frac{\Psi_{kL}}{\Lambda_L} - \tilde{\lambda}_k [\Phi(\boldsymbol{\sigma})]_{L,k} \right) d \log M_k. \end{aligned} \tag{B-15}$$

Finally, substitute the wedge relation (10), $d\log\mu_k = -d\log\tau_k - \xi_k\, d\log M_k$, to obtain

$$d\log Y = -\sum_{k=1}^{N} \tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k}\, d\log\tau_k + \sum_{k=1}^{N}\left[\eta_k\lambda_k\frac{\Psi_{kL}}{\Lambda_L} - (\eta_k+\xi_k)\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma})]_{L,k}\right] d\log M_k. \tag{B-16}$$

The coefficients on $d\log\tau_k$ and $d\log M_k$ are the partial derivatives stated in the proposition. Combining them with the equilibrium entry response $d\log M_l/d\log\tau_k$ gives the total derivative and the entry-centrality decomposition. □

Proof of Proposition 2. The entry response is pinned down by three log-linear relations: the wedge relation (10), the free-entry relation (11), and the sales-share equation (12). The free-entry relation gives

$$d\log\boldsymbol{\lambda} = \left(I + (D_\varepsilon - I)D_\xi\right) d\log\boldsymbol{M},$$

where $D_\varepsilon = \mathrm{diag}(\boldsymbol{\varepsilon})$ and $D_\xi = \mathrm{diag}(\boldsymbol{\xi})$. The sales-share equation can be written compactly as

$$d\log\boldsymbol{\lambda} = -\Lambda_\mu\, d\log\boldsymbol{\mu} + \Lambda_M\, d\log\boldsymbol{M},$$

where

$$[\Lambda_\mu]_{l,k} = \lambda_k(\Psi_{kl} - \delta_{kl})/\lambda_l + [\Phi(\boldsymbol{\sigma}-\mathbf{1})]_{l,k}\tilde{\lambda}_k, \qquad \Lambda_M = \Phi(\boldsymbol{\sigma}-\mathbf{1})\,\mathrm{diag}(\tilde{\boldsymbol{\lambda}}\circ\boldsymbol{\eta}).$$

Using $d\log\boldsymbol{\mu} = -d\log\boldsymbol{\tau} - D_\xi d\log\boldsymbol{M}$ in the sales-share equation and equating it to the free-entry relation gives

$$\left(I + (D_\varepsilon - I)D_\xi\right) d\log\boldsymbol{M} = \Lambda_\mu\, d\log\boldsymbol{\tau} + (\Lambda_M + \Lambda_\mu D_\xi)\, d\log\boldsymbol{M}, \tag{B-17}$$

so collecting the $d\log\boldsymbol{M}$ terms gives

$$\frac{d\log\boldsymbol{M}}{d\log\boldsymbol{\tau}} = \left[I - \Lambda_M + (D_\varepsilon - I - \Lambda_\mu)D_\xi\right]^{-1}\Lambda_\mu. \tag{B-18}$$

The derivative on the left is the matrix whose (l,k) element is $\partial\log M_l/\partial\log\tau_k$. At $\boldsymbol{\xi} = \mathbf{0}$, the D_ξ term vanishes, giving the Dixit–Stiglitz cascade $(I - \Lambda_M)^{-1}\Lambda_\mu$. □

Proof of Corollary 2. Under Cobb–Douglas production and consumption, $\boldsymbol{\sigma} = \mathbf{1}$. Lemma 1 then gives

$$[\Phi(\mathbf{1})]_{L,k} = \frac{\lambda_k}{\tilde{\lambda}_k}\frac{\Psi_{kL}}{\Lambda_L} - 1.$$

Substituting this into Proposition 1 yields

$$\frac{\partial\log Y}{\partial\log\tau_k} = -\tilde{\lambda}_k[\Phi(\mathbf{1})]_{L,k} = \tilde{\lambda}_k - \lambda_k\frac{\Psi_{kL}}{\Lambda_L}.$$

Since Dixit–Stiglitz demand sets $\xi_k = 0$ and $\eta_k = 1/(\varepsilon_k - 1)$,

$$\frac{\partial \log Y}{\partial \log M_l} = \eta_l \lambda_l \frac{\Psi_{lL}}{\Lambda_L} - \eta_l \tilde{\lambda}_l [\Phi(\mathbf{1})]_{L,l}$$
$$= \eta_l \tilde{\lambda}_l = \frac{\tilde{\lambda}_l}{\varepsilon_l - 1}.$$

Finally, $\boldsymbol{\sigma} = \mathbf{1}$ implies $\Lambda_M = \mathbf{0}$, and $\boldsymbol{\xi} = \mathbf{0}$ implies that Proposition 2 reduces to

$$\frac{d \log \boldsymbol{M}}{d \log \boldsymbol{\tau}} = \Lambda_\mu.$$

Since $\Phi(\boldsymbol{\sigma} - \mathbf{1}) = \Phi(\mathbf{0}) = 0$ in the Cobb–Douglas case,

$$[\Lambda_\mu]_{l,k} = \frac{\lambda_k}{\lambda_l}(\Psi_{kl} - \delta_{kl}).$$

Combining these three objects gives the decomposition stated in the corollary. □

Derivation for the vertical economy. Each sector uses only the good immediately upstream and the household consumes only sector 1, so the chain factorizes:

$$\Psi_{1L} = \Psi_{1k}\Psi_{kL} \qquad \text{for every } k.$$

Hence $\lambda_k \Psi_{kL}/\Lambda_L = 1$. The absence of within-sector input substitution makes $[\Phi(\boldsymbol{\sigma})]_{L,k} = 0$. Proposition 1 then gives

$$\frac{\partial \log Y}{\partial \log \tau_k} = 0, \qquad \frac{\partial \log Y}{\partial \log M_j} = \eta_j.$$

Forward propagation is inactive, $\Lambda_M = \mathbf{0}$, and along the chain the sales-share matrix reduces to $[\Lambda_\mu]_{jl} = \lambda_l \Psi_{lj}/\lambda_j - \delta_{jl} = \mathbf{1}(l < j)$. Writing $m_{jk} \equiv \partial \log M_j / \partial \log \tau_k$ for the entry response, collected in the matrix $\boldsymbol{m}$, and $\delta_j^{-1} = 1 + (\varepsilon_j - 1)\xi_j$, the cascade of Proposition 2, $\big(I + (D_\varepsilon - I - \Lambda_\mu)D_\xi\big)\boldsymbol{m} = \Lambda_\mu$, reads row by row as

$$\delta_j^{-1} m_{jk} = \mathbf{1}(j > k) + \sum_{k<l<j} \xi_l m_{lk}. \tag{B-19}$$

Solving recursively from downstream to upstream gives $m_{jk} = 0$ for $j \le k$ and, for $j > k$,

$$\frac{\partial \log M_j}{\partial \log \tau_k} = \delta_j \prod_{k<h<j} \big(1 + \xi_h \delta_h\big), \qquad \delta_k = \frac{1}{1 + (\varepsilon_k - 1)\xi_k}, \tag{B-20}$$

as stated in the text. When $\xi_k = 0$ each $\delta_k = 1$ and the response collapses to the one-for-one upstream cascade $\mathbf{1}(j > k)$. □

Derivation for the horizontal economy. With no intermediate inputs, the cost- and revenue-based

Leontief inverses satisfy $\tilde{\Psi}_{(k)} = e_k$ and $\Psi_{(l)} = e_l$ on the sector block, while $\Psi_{lL} = \mu_l^{-1}$ and $\lambda_l = \tilde{\lambda}_l = b_l$. Evaluating the covariance operator at household expenditure weights gives

$$[\Phi(\boldsymbol{\sigma})]_{L,k} = \sigma_0 \mu_k^{-1}(\bar{\mu} - \mu_k), \qquad [\Phi(\boldsymbol{\sigma})]_{lk} = \sigma_0(\delta_{kl}/b_l - 1),$$

where $\Lambda_L = \mathbb{E}_b(\mu^{-1})$ and $\bar{\mu} = \Lambda_L^{-1}$. Proposition 1 then yields the distortion centrality and

$$\begin{aligned} \frac{\partial \log Y}{\partial \log M_l} &= \eta_l b_l \frac{\bar{\mu}}{\mu_l} + \sigma_0(\eta_l + \xi_l) b_l \left(1 - \frac{\bar{\mu}}{\mu_l}\right) \\ &= \eta_l b_l + \Big[(\sigma_0 - 1)\eta_l + \sigma_0 \xi_l\Big] b_l \mu_l^{-1}(\mu_l - \bar{\mu}). \end{aligned} \tag{B-21}$$

The remaining step is to compute the entry response. On the sector block,

$$\Lambda_\mu = (1 - \sigma_0)(\mathbf{1}\boldsymbol{b}' - I), \qquad \Lambda_M = (1 - \sigma_0)(\mathbf{1}\boldsymbol{b}' - I)\mathrm{diag}(\boldsymbol{\eta}).$$

The cascade operator in Proposition 2 is therefore

$$A \equiv I - \Lambda_M + \Big(D_\varepsilon - I - \Lambda_\mu\Big) D_\xi = \mathrm{diag}(\tilde{\boldsymbol{\chi}}^{-1}) - (1 - \sigma_0)\,\mathbf{1}\Big(\boldsymbol{b} \circ (\boldsymbol{\eta} + \boldsymbol{\xi})\Big)', \tag{B-22}$$

with $\tilde{\chi}_k^{-1} = 1 + (1 - \sigma_0)\eta_k + (\varepsilon_k - \sigma_0)\xi_k$. Define $\omega_k = \tilde{\chi}_k[1 + (\varepsilon_k - 1)\xi_k]$. The identity $(1 - \sigma_0)(\eta_k + \xi_k)\tilde{\chi}_k = 1 - \omega_k$ implies

$$1 - (1 - \sigma_0)\mathbb{E}_b[(\eta + \xi)\tilde{\chi}] = \mathbb{E}_b(\omega).$$

Applying the Sherman–Morrison formula gives

$$A^{-1} = \mathrm{diag}(\tilde{\boldsymbol{\chi}}) + \frac{1 - \sigma_0}{\mathbb{E}_b(\omega)}\,\tilde{\boldsymbol{\chi}}\Big(\boldsymbol{b} \circ (\boldsymbol{\eta} + \boldsymbol{\xi}) \circ \tilde{\boldsymbol{\chi}}\Big)'. \tag{B-23}$$

Multiplying by Λ_μ and simplifying,

$$\frac{\partial \log M_l}{\partial \log \tau_k} = (1 - \sigma_0)\,\tilde{\chi}_l\Big(b_k \frac{\omega_k}{\mathbb{E}_b(\omega)} - \delta_{kl}\Big). \tag{B-24}$$

With homogeneous wedges the output response to entry is $\partial \log Y / \partial \log M_l = \eta_l b_l$, so

$$\frac{d \log Y}{d \log \tau_k} = \sum_{l=1}^{N} \eta_l b_l \frac{\partial \log M_l}{\partial \log \tau_k} = (1 - \sigma_0)\, b_k \tilde{\chi}_k \Big[\Big(1 + (\varepsilon_k - 1)\xi_k\Big)\frac{\mathbb{E}_b(\eta\tilde{\chi})}{\mathbb{E}_b(\omega)} - \eta_k\Big], \tag{B-25}$$

which is the entry centrality in Corollary 3; the distortion centrality vanishes because wedges are homogeneous. Setting $\xi_k = 0$ gives $\tilde{\chi}_k = \omega_k = \chi_k$ and, using $\mathbb{E}_b(\chi) = 1 - (1 - \sigma_0)\mathbb{E}_b(\eta\chi)$, the Dixit–Stiglitz expression $(1 - \sigma_0) b_k \chi_k [\mathbb{E}_{b\chi}(\eta) - \eta_k]$. □

Derivation for the roundabout economy. In the one-sector economy, the covariance matrix can be evaluated directly. The relevant substitution matrix is

$$\Phi(1) = \begin{bmatrix} \mu_1^{-1}\lambda_1\tilde{\lambda}_1^{-1}(1-\tilde{\lambda}_1^{-1}) & 0 \\ -\lambda_1(1-\tilde{\lambda}_1^{-1})(1-\mu_1^{-1}) & 0 \end{bmatrix} \tag{B-26}$$

so the labor-row element under elasticity σ_1 is

$$[\Phi(\boldsymbol{\sigma})]_{L,1} = -\sigma_1\lambda_1(1-\tilde{\lambda}_1^{-1})(1-\mu_1^{-1}).$$

At the no-subsidy benchmark, $\mu_1 = \varepsilon_1/(\varepsilon_1 - 1)$, so $1-\mu_1^{-1} = 1/\varepsilon_1$. Proposition 1 then gives the intensive-margin effect:

$$\frac{\partial \log Y}{\partial \log \tau_1} = -\tilde{\lambda}_1[\Phi(\boldsymbol{\sigma})]_{L,1} = \sigma_1\lambda_1(\tilde{\lambda}_1 - 1)/\varepsilon_1, \tag{B-27}$$

which is the distortion-centrality expression in the text. The same proposition gives the output value of entry. Since $\lambda_1\Psi_{1L}/\Lambda_L = \lambda_1$ in the one-sector economy,

$$\frac{\partial \log Y}{\partial \log M_1} = \eta_1\lambda_1 - (\eta_1+\xi_1)\tilde{\lambda}_1[\Phi(\boldsymbol{\sigma})]_{L,1} \tag{B-28}$$

$$= \eta_1\lambda_1 + (\eta_1+\xi_1)\sigma_1\lambda_1(\tilde{\lambda}_1-1)/\varepsilon_1. \tag{B-29}$$

It remains to compute the entry response. In the scalar case,

$$(1-\Lambda_M)^{-1} = \left(1-(\sigma_1-1)\Phi^1(1)\tilde{\lambda}_1\eta_1\right)^{-1} = [1-\eta_1(\sigma_1-1)(\lambda_1-1)]^{-1} \tag{B-30}$$

and

$$\Lambda_\mu = \sigma_1(\lambda_1 - 1). \tag{B-31}$$

Thus $\Lambda_M = \eta_1(\sigma_1-1)(\lambda_1-1)$, and Proposition 2 gives

$$\begin{aligned} \frac{d\log M_1}{d\log \tau_1} &= \frac{\Lambda_\mu}{1-\Lambda_M+(\varepsilon_1-1-\Lambda_\mu)\xi_1} \\ &= \frac{\sigma_1(\lambda_1-1)}{1+\eta_1(1-\sigma_1)(\lambda_1-1)+\xi_1\left[(\varepsilon_1-1)-\sigma_1(\lambda_1-1)\right]}. \end{aligned} \tag{B-32}$$

□

C Industrial Policy Through Entry Costs

The main text studies sectoral subsidies that lower variable wedges and thereby affect entry through sectoral sales. Some industrial policies operate more directly on the extensive margin by lowering the fixed cost of operation or entry. Examples include start-up grants, licensing reform, subsidized credit for new establishments, industrial-park access, and policies that reduce compliance or set-up costs. This appendix shows that such policies enter the same sufficient-statistic framework through a different entry impulse.

Let κ_k denote an entry-cost policy in sector k, normalized so that a higher κ_k lowers the effective fixed cost from f_k to f_k/κ_k. The zero-profit condition becomes

$$M_k = \frac{1}{\varepsilon_k} \frac{P_k Y_k}{f_k/\kappa_k} = \frac{\kappa_k \lambda_k}{\varepsilon_k \phi_k}. \tag{C-1}$$

Log-linearizing around the no-policy benchmark gives

$$\Big(1 + (\varepsilon_k - 1)\xi_k\Big) d\log M_k = d\log\lambda_k + d\log\kappa_k. \tag{C-2}$$

Relative to equation (11), a fixed-cost subsidy creates a direct entry impulse even holding sectoral sales fixed. Intuitively, reducing the set-up cost allows more firms to break even at a given market size.

The sales-share equation is unchanged:

$$d\log\boldsymbol{\lambda} = -\Lambda_\mu\, d\log\boldsymbol{\mu} + \Lambda_M\, d\log\boldsymbol{M}, \tag{C-3}$$

and the wedge relation remains

$$d\log\boldsymbol{\mu} = -d\log\boldsymbol{\tau} - D_\xi d\log\boldsymbol{M}.$$

Combining these equations with (C-2) yields

$$\frac{d\log\boldsymbol{M}}{d\log\boldsymbol{\kappa}} = \Big[I - \Lambda_M + (D_\varepsilon - I - \Lambda_\mu)D_\xi\Big]^{-1}, \tag{C-4}$$

where the derivative is the matrix whose (l,k) element is $\partial\log M_l/\partial\log\kappa_k$. Thus an entry-cost subsidy has the same propagation operator as the output subsidy in Proposition 2, but a different impulse. An output subsidy first changes wedges and sales shares, so its impulse is Λ_μ; a fixed-cost subsidy directly raises the mass of entrants in the targeted sector, so its impulse is the identity matrix.

The resulting output effect follows immediately from Proposition 1. Since $\boldsymbol{\kappa}$ does not directly change variable wedges, its first-order effect on output operates through induced entry:

$$\frac{d\log Y}{d\log\kappa_k} = \sum_{l=1}^{N}\left[\eta_l\lambda_l\frac{\Psi_{lL}}{\Lambda_L} - (\eta_l + \xi_l)\tilde{\lambda}_l[\Phi(\boldsymbol{\sigma})]_{L,l}\right]\frac{\partial\log M_l}{\partial\log\kappa_k}. \tag{C-5}$$

Equation (C-5) is the fixed-cost-policy analogue of entry centrality. It ranks sectors by the output value of lowering entry costs in sector k, including both the direct entrants created in that sector and the entry cascade transmitted through the production network.

In the Cobb–Douglas, Dixit–Stiglitz benchmark, $\Lambda_M = \mathbf{0}$ and $D_\xi = \mathbf{0}$, so (C-4) reduces to

$$\frac{d\log \boldsymbol{M}}{d\log \boldsymbol{\kappa}} = I.$$

Fixed-cost policies then generate only direct entry in the targeted sector, and

$$\frac{d\log Y}{d\log \kappa_k} = \frac{\tilde{\lambda}_k}{\varepsilon_k - 1}.$$

With non-Cobb–Douglas substitution or variable markups, however, the direct entry induced by a fixed-cost policy changes relative prices and wedges, reallocates sales, and triggers further entry responses across sectors through the same cascade operator in (C-4).

This extension also clarifies the interpretation of the main text. Output subsidies and fixed-cost subsidies both work through the extensive margin, but they act on different terms in firms' entry condition. Output subsidies expand entry by changing wedges and sales shares; fixed-cost subsidies expand entry by lowering the break-even scale. The output value of either policy is governed by the same sector-level value of entry, but the network impulse that starts the cascade differs.

D Overhead Paid in Labor

The main text treats the overhead f_k as a profit claim that is rebated lump-sum, so that the extensive margin does not absorb resources (Section 2). This isolates the variety and pro-competitive channels but removes the resource cost of creating firms. We now develop the alternative, standard in much of the literature, in which the overhead is paid in labor and competes with production for the scarce factor, as in Dixit and Stiglitz (1977), Krugman (1979), Melitz (2003), Baqaee (2018), and Baqaee and Farhi (2020a).

Environment and equilibrium. Each operating firm in sector k now requires $f_k > 0$ units of labor to cover its overhead, in addition to the labor and intermediates used in variable production. The variable cost-minimization problem and the pricing rule $p_k(\nu) = \mu_k \, mc_k$ are unchanged; what changes is that overhead is now paid in labor rather than rebated as a transfer.

A firm's profit is its revenue net of the subsidized variable cost and the overhead labor cost,

$$\pi_k(\nu) = p_k(\nu)\, y_k(\nu) - \tau_k^{-1} mc_k \, y_k(\nu) - W f_k, \tag{D-1}$$

and the household budget constraint becomes

$$\sum_{k=1}^{N} P_k C_k = W\bar{L} + \sum_{k=1}^{N} \int_0^{M_k} \pi_k(\nu)\, d\nu - T, \tag{D-2}$$

with the overhead now paid out as wages. The labor market clears against production and overhead, $L + \sum_{k=1}^{N} M_k f_k = \bar{L}$, where $L = \sum_{k=1}^{N} L_k$ is aggregate production labor and $L_k = \int_0^{M_k} l_k(v)dv$ is sector k's production labor.

Because the variable production block and the input–output network are unchanged, equations (B-1) to (B-4) and (B-8) to (B-13) carry over verbatim. The aggregation theorem, however, gains one term: with production labor L no longer equal to the fixed endowment, the bridge from labor income to output retains a $d\log L$,

$$d\log Y = \sum_{k=1}^{N} \tilde{\lambda}_k \Big(d\log z_k + \eta_k \, d\log M_k - d\log \mu_k \Big) - d\log \Lambda_L + d\log L. \tag{D-3}$$

Here Λ_L is the income share of production labor L—the part of value added accruing to labor employed in variable production, as distinct from the overhead labor $\sum_{k=1}^{N} M_k f_k$. The labor-market clearing and free-entry conditions together pin the response of production labor,

$$d\log L = -\sum_{k=1}^{N} \frac{\lambda_k}{\varepsilon_k \Lambda_L} \, d\log M_k, \tag{D-4}$$

so (D-3) can be rewritten as

$$\begin{aligned} d\log Y &= \sum_{k=1}^{N} \tilde{\lambda}_k \Big(d\log z_k + \eta_k \, d\log M_k - d\log \mu_k \Big) - d\log \Lambda_L - \sum_{k=1}^{N} \frac{\lambda_k}{\varepsilon_k \Lambda_L} \, d\log M_k \\ &= \sum_{k=1}^{N} \tilde{\lambda}_k \, d\log z_k + \sum_{k=1}^{N} \Big(\tilde{\lambda}_k \eta_k - \frac{\lambda_k}{\varepsilon_k \Lambda_L} \Big) d\log M_k - \sum_{k=1}^{N} \tilde{\lambda}_k \, d\log \mu_k - d\log \Lambda_L. \end{aligned} \tag{D-5}$$

The value of an entrant in sector k is therefore the variety gain net of the labor it pulls into overhead, $\tilde{\lambda}_k \eta_k - \lambda_k / (\varepsilon_k \Lambda_L)$, in place of the rebated economy's uncosted gain $\tilde{\lambda}_k \eta_k$.

In this setup the aggregation results of the main text each carry over with a single change, so we state them in place rather than as separate theorems. The output decomposition of Theorem 1, equation (5), is now equation (D-5): the technology term $\sum_{k=1}^{N} \tilde{\lambda}_k \, d\log z_k$ and the allocative-efficiency term $-\sum_{k=1}^{N} \tilde{\lambda}_k \, d\log \mu_k - d\log \Lambda_L$ are unchanged, and only the variety coefficient changes, from the rebated economy's uncosted gain $\tilde{\lambda}_k \eta_k$ to the net value $\tilde{\lambda}_k \eta_k - \frac{\lambda_k}{\varepsilon_k \Lambda_L}$. The change carries a normative bite the baseline lacks: the rebated coefficient is always positive, whereas the net value can take either sign—positive when returns to variety are large relative to the overhead an entrant absorbs, negative otherwise—and vanishes at efficient entry, the isolated-sector Dixit–Stiglitz point $\mu_k = \eta_k \varepsilon_k$ that input–output linkages generally shift.

Specializing to sectoral subsidies, the two-margin decomposition (6) is now

$$d\log Y = \underbrace{\sum_{k=1}^{N}\left(\tilde{\lambda}_k + \frac{d\log\Lambda_L}{d\log\mu_k}\right)d\log\tau_k}_{\text{intensive margin}} \\ + \underbrace{\sum_{k=1}^{N}\left[\eta_k\tilde{\lambda}_k - \frac{\lambda_k}{\varepsilon_k\Lambda_L} + \xi_k\left(\tilde{\lambda}_k + \frac{d\log\Lambda_L}{d\log\mu_k}\right) - \frac{d\log\Lambda_L}{d\log M_k}\right]d\log M_k}_{\text{extensive margin}}, \tag{D-6}$$

which coincides with its rebated analogue (6) except that the extensive margin loads on the net variety value $\eta_k\tilde{\lambda}_k - \frac{\lambda_k}{\varepsilon_k\Lambda_L}$ in place of $\eta_k\tilde{\lambda}_k$.

Likewise the entry cascade of Proposition 2, equation (13), is now

$$\frac{d\log \boldsymbol{M}}{d\log\boldsymbol{\tau}} = \left[I - \Lambda_M + \left(D_\varepsilon - I - \Lambda_\mu\right)D_\xi + \mathbf{1}\boldsymbol{w}_M'\right]^{-1}\left(\Lambda_\mu - \mathbf{1}\boldsymbol{w}_\mu'\right), \tag{D-7}$$

with the same network operator and wedge impulse Λ_μ as in Proposition 2. The new rank-one terms $\mathbf{1}\boldsymbol{w}_M'$ and $-\mathbf{1}\boldsymbol{w}_\mu'$ are the general-equilibrium wage feedback the rebated model nets away, built from the labor-income-share response to wedges,

$$w_{\mu,k} \equiv -\frac{\partial\log\Lambda_L}{\partial\log\mu_k} = \tilde{\lambda}_k\Big(1 + [\Phi(\boldsymbol{\sigma})]_{L,k}\Big),$$

and the response of the wage feedback $d\log\Lambda_L - d\log L$ to entry,

$$w_{M,k} = \eta_k\tilde{\lambda}_k[\Phi(\boldsymbol{\sigma} - \mathbf{1})]_{L,k} + \xi_k\tilde{\lambda}_k\Big(1 + [\Phi(\boldsymbol{\sigma})]_{L,k}\Big) + \frac{\lambda_k}{\varepsilon_k\Lambda_L}.$$

Because overhead is now paid in labor, entry anywhere draws labor into overhead and bids up the wage, raising every sector's break-even scale and crowding out entry economy-wide; since this feedback runs through the single scalar Λ_L, it enters as a uniform shock $\mathbf{1}$ propagated by the same multiplier, and by the Sherman–Morrison identity the cascade is the rebated multiplier of Proposition 2 plus a rank-one tail. Setting $\boldsymbol{w}_\mu = \boldsymbol{w}_M = \mathbf{0}$ recovers that cascade term for term. The comparison is sharpest in the constant-wedge, Cobb–Douglas benchmark ($\boldsymbol{\xi} = \mathbf{0}$, $\Lambda_M = \mathbf{0}$): the rebated cascade is the bare sales-share impulse, $d\log M_l/d\log\tau_k = \frac{\lambda_k}{\lambda_l}(\Psi_{kl} - \delta_{kl})$, whereas here it is shifted down by the uniform wage feedback $\frac{\lambda_k}{\mu_k}$, $d\log M_l/d\log\tau_k = \frac{\lambda_k}{\lambda_l}(\Psi_{kl} - \delta_{kl}) - \frac{\lambda_k}{\mu_k}$.